

Project Report Title

AI-BASED DETECTION OF HUMAN EMOTIONS THROUGH MULTIMODAL DATA FUSION

By

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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Abstract

In the field of emotion recognition, Artificial Intelligence (AI) has made great strides in interpreting emotional states from various forms of data, leading to the development of intelligent systems. AI has paved the way for significant developments in human emotion recognition, with the ability to interpret emotional states from various forms of data, such as the emergence of intelligent systems. Typical unimodal emotion recognition systems based on a single modality (such as face or speech) are limited in recognising emotions in real-world scenarios because of changes in lighting, background noise and the lack of emotive signals. This research introduces a multimodal emotion recognition system leveraging AI that combines facial images with speech to enhance the overall classification accuracy and robustness. Model development includes an image preprocessing technique, in which the images are resized and preprocessed by data augmentation and an audio preprocessing technique, in which the audio is preprocessed by normalisation and a feature extraction technique. Three unimodal deep learning models, namely a Custom Convolutional Neural Network (CNN), an EfficientNetB0 transfer learning model and an Audio CNN-BiLSTM model, are implemented and evaluated.

A multimodal fusion framework is generated by fusing the facial and audio feature embeddings to classify emotions. Experimental results show that the proposed multimodal fusion framework outperforms unimodal models (both in general) on all the evaluation metrics. The performance evaluation results of the multimodal model show high accuracy (96.82%), high precision (96.76%), high recall (96.92%), high F1 score (96.81%), and high ROC-AUC (99.77%), which indicate that the emotion recognition performance of the multimodal model has also improved significantly. The comparative analysis shows that a multimodal approach yields better robustness, improves classification error reduction and predictive capability than unimodal approaches. The results show the efficiency of multimodal deep learning to automatically recognise human emotions and reveal its potential role in a variety of fields such as healthcare, intelligent tutoring systems, customer service, social robotics, and human–computer interaction.

Results of the experimental evaluation showed that the proposed multimodal framework always outperformed the individual unimodal models. The unimodal architectures were able to infer some kind of interpretable visual and auditory features, but each model showed its limits stemming from having only one source of information of emotions. To overcome these challenges, the multimodal fusion approach was employed to combine complementary facial and speech

representations and made substantial contributions to the classification accuracy. Comparative analysis also showed that multimodal learning was more effective in lowering the classification errors and was more robust as compared to unimodal learning.

This research reveals the potential of multimodal deep learning in automated human emotion recognition better than unimodal. The proposed system integrated facial and speech data into an integrated AI system, which resulted in improved reliability, predicability, and robustness to the usual variations in real-world indices. Thus the proposed framework can be successfully applied to different real world applications such as emotion-aware human–computer interaction, emotionaware natural language processing, smart assistants, intelligent learning environments, smart healthcare monitoring, emotion-aware customer service systems, and emotion-aware social robots. Future work could be extended to include more modalities like textual data and physiological signals, more complex and large sets of data, and computational optimisation, to be deployed to real-time intelligent systems.

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Chapter 1 Introduction

1.1 Problem Description, Context and Motivation

The rise of artificial intelligence and deep learning technology has made a big breakthrough, there are still difficulties in recognising human emotions accurately. Current emotion recognition systems are very limited and only use one of the following modalities: facial expression, speech signal, or textual information. These unimodal methods are effective in controlled conditions, but are less effective in real-world settings where these can fail to achieve good accuracy. Poor lighting conditions, occlusion of the face, background noise, language variability and fuzzy meanings of words in the text can make a big difference in their reliability and robustness. The expression of emotions is multi-layered and complex [1]. Moreover, various proposed models are tested with benchmark datasets which are not necessarily representative of real-world scenarios, and this reduces their applicability to real life.

1.2 Research Aim and Objectives

Research Aim

The aim of this project is to design, develop, and test an AI-based, multimodal emotion recognition framework that combines facial, speech and text-based features to enhance the accuracy, stability and consistency of human emotion recognition as compared to the traditional unimodal programs.

Research Objectives

- To critically analyse the existing artificial intelligence and deep learning approaches to human emotion recognition in any data form.
- To examine existing multimodal data fusion methods and to determine their advantages, weaknesses and their applicability to emotion detection tasks.
- To model and deploy an AI system based on multiple languages, characteristics, and data sets (facial, speech, and text) to identify emotions.
- To test the framework proposed on benchmark datasets that are publicly available and proper performance indicators.
- To compare the performance of the presented multimodal with individual unimodal models by classification accuracy, robustness, and overall performance.

- To determine possible challenges and practical aspects related to the implementation of multimodal emotion recognition systems in the real world.

1.3 Methodology

In this study, quantitative research will be applied for obtaining and testing the proposed Artificial Intelligence (AI) based Multiple Modal Emotion Recognition System (MRS). Qualitative research will be deemed appropriate since the main purpose of the research is to evaluate the classification performance of these different deep learning models using numerical evaluation. This study will not examine mere subjective opinions or personal experiences, but instead use machine learning algorithms to produce measurable results. Objective performance measures will be employed to test the efficacy of the suggested framework, such as accuracy, precision, recall, F1-score and Receiver Operating Characteristic-Area Under the Curve (ROC-AUC) [2]. This study follows the experimental research method, a method used to study whether an Artificial Intelligence (AI) based multimodal framework for human emotion recognition is effective or not.

The experimental strategy will be appropriate: The research shall be conducted in a controlled experimental environment to develop, train and evaluate deep learning models based on benchmark datasets. This approach will allow intentional testing of the unimodal and multimodal approaches and allow comparison of unimodal to multimodal approaches, while allowing for objective measurement of differences in approach performance. The research approach will be the use of various deep learning architectures by means of various data modalities [3]. Facial emotion recognition will be used with the aid of a Custom Convolutional Neural Network (CNN) and an EfficientNetB0 transfer learning model, while the emotion classification of voice signals will be performed with the aid of a CNN-BiLSTM model.

1.4 Legal, Social, Ethical and Professional Considerations

Ethical and Legal Issues

As seen above, the project uses multimodal data which include facial expressions, speech signals and other types of texts that may have personal and biometric data. It will thus be important to ensure that privacy is observed while handling this type of information. This will be achieved by ensuring informed consent and license of use of human subjects or open source data sets, whereby the data is de-identified and used only for research purposes.

Handling and processing of data in the project will follow the applicable laws including GDPR. Appropriate security measures such as encryption will reduce risks involved in handling sensitive information.

Social and Professional Implications

The emotional detection AI raises socially and professionally significant questions aside from those concerning efficiency [4]. Whether male or female, young or old, Caucasian or Asian or another ethnicity, English or Russian or any other language – all machine learning algorithms can be biased despite being based on balanced or unbalanced data sets, and therefore generate biased results. This problem can be addressed using several different data sets and evaluations.

In this scenario, REA can assist in the appropriate use of artificial intelligence, which encompasses transparency and accountability. The users need validation and oversight from the developers concerning the abilities of the automated emotion recognition systems.

1.5 Background

Artificial Intelligence (AI) has revolutionised the way intelligent systems perceive, interpret, and respond to human behaviour in many applications. Affective computing is one of the swiftest emerging domains of AI, emphasising the understanding of human emotions. The recognition of emotions is more important than ever because emotions have a strong and significant impact on communication, decision-making, learning, and social interactions. Accurate emotion detection can enable emotion-adaptive human-computer interaction, leading to more individualised and contextdependent systems, which improve the quality of human interaction. Moreover, facial, speech, and textual clues all contain their own unique emotion signals, and to overcome these problems, the task of emotion recognition has increasingly begun to focus on multimodal emotion recognition problems addressing how facial, speech and textual information can be combined to extract complementary emotion signals [5]. Another significant development in multimodal learning has been the adoption of deep learning techniques, such as CNN, RNN, and transformer models, which automatically learn meaningful representations from multimodal data.

1.6 Structure of Report

Chapter 1 provides an overview of the research problem, objectives, method, ethical issues, and project background. Chapter 2 highlights relevant literature and technologies in relation to emotion recognition using artificial intelligence and data fusion. Chapter 3 describes the process of

implementing the proposed multimodal framework involving data processing, model development, and framework structure. The performance of the proposed model is presented and compared with other models in Chapter 4. The final chapter, Chapter 5, includes a summary of the research findings and future directions of the research.

Chapter 2 Literature- Technology Review

2.1 Literature Review

2.1.1 Artificial Intelligence and Human Emotion Recognition

As discussed by Chandraumakantham *et al.* (2024), Artificial Intelligence (AI) is one of the most promising technologies of the twenty-first century that mimics human intelligence to serve its goals like learning, reasoning, perception, problem solving and decision making [6]. Thanks to recent breakthroughs in computational power, volume of data at hand, and deep learning algorithms, AI's applications are rapidly increasing across such fields as healthcare, education, finance, security, and intelligent human-computer interaction. Emotion recognition is one of the fastest-growing subfields of AI that seeks to enable machines to learn how to recognise, interpret, and react to people's emotions. The sensitivity of human emotion has been considered an important issue in affective computing due to the effect of emotions on communication, behaviour, the decision-making process and interpersonal relations.

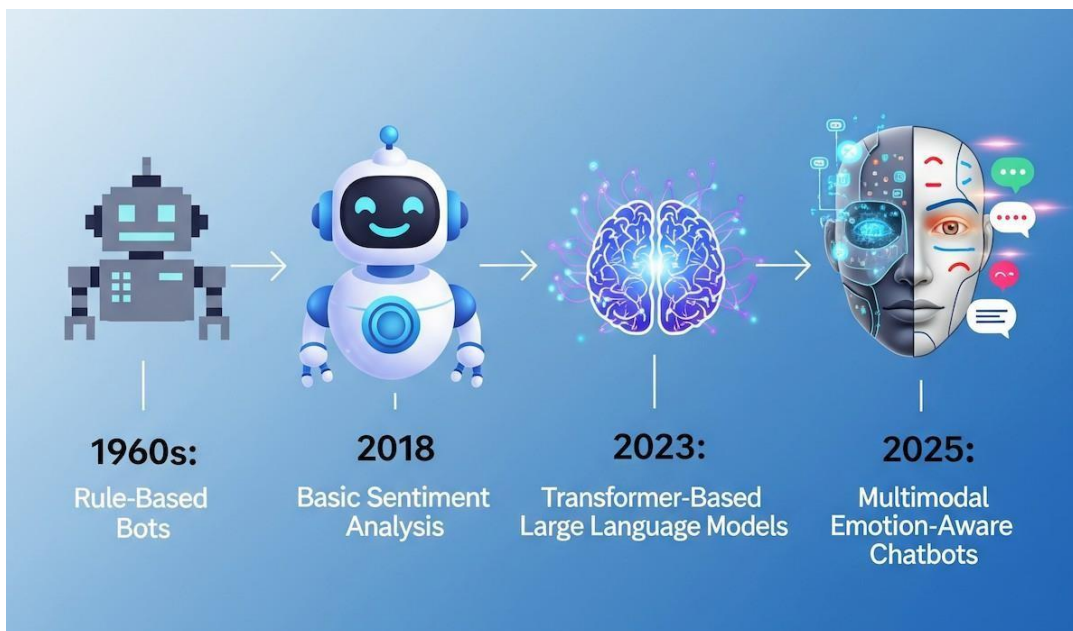


Figure 2.1.1.1: Artificial Intelligence and Human Emotion Recognition

(Source: Cascella *et al.* 2024)[7]

The first generation of AI-based emotion recognition models has been based on classic machine learning and took handcrafted features from the facial expressions, speech signals or textual information and applied some sort of classification technique on top. Some of the algorithms,

including Support Vector Machines (SVM), Decision Trees, Random Forests and k-Nearest Neighbour (k-NN), have been found to have good performance in controlled settings [8]. Both, however, relied heavily on manual feature engineering, and struggled to generalise to different datasets and real-world applications.

The latest advancements in the field of Artificial Intelligence have further contributed to the improvement of emotion recognition due to the incorporation of deep learning, cloud computing, as well as edge AI technologies. The existing AI models can generate meaningful responses to multimodal data in large amounts in real time and facilitate more responsive and adaptive humancomputer interaction. Applications of AI-based emotion recognition are ever growing in mental healthcare, driver monitoring systems, virtual assistants and smart learning environments [46]. Although these advances have been made the main challenges include making sure that users have privacy, decreasing algorithmic bias and enhancing the transparency of AI decisions. The needs to attend to these issues are critical in ensuring ethical and responsible implementation of emotionsensitive intelligent systems.

2.1.2 Theoretical Foundations of Affective Computing and Emotional Intelligence

As reported by Gutierrez *et al.* (2024), Artificial Intelligence (AI) emotion recognition systems are based on the principles of affective computing and emotional intelligence that enable the identification, processing and management of emotions in human interactions [9]. Theories of emotion serve as the basis for the design of systems that are able to detect the emotional state from various types of information.

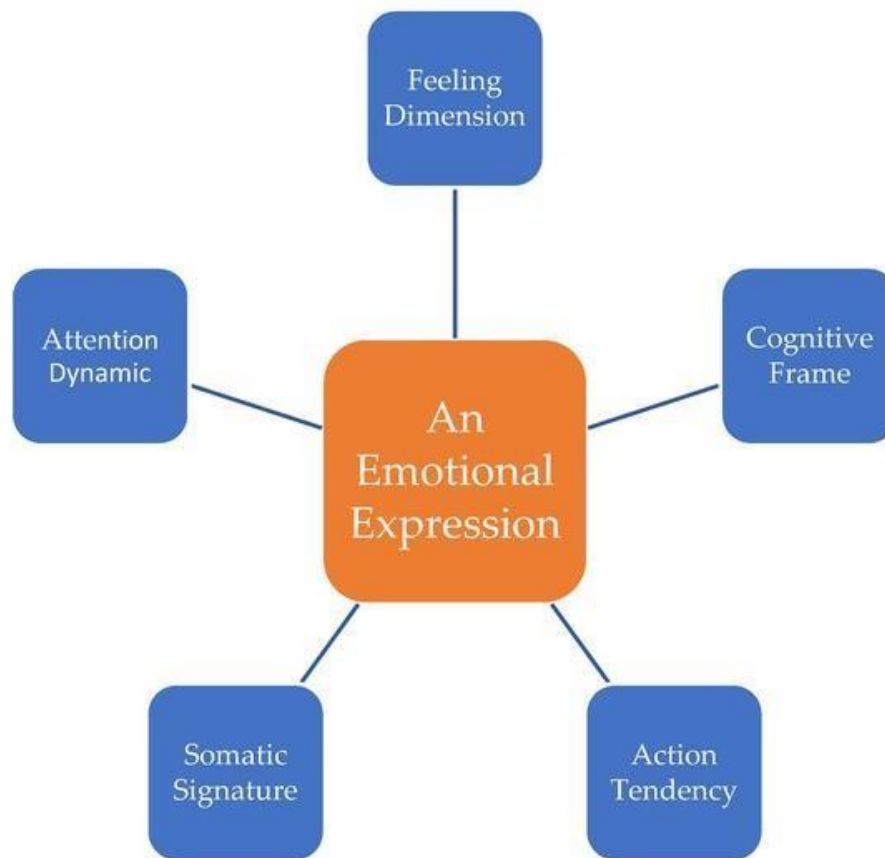


Figure 2.1.2.1: Theoretical Foundations of Affective Computing and Emotional Intelligence

(Source: Zhu, 2024), [10]

Affective computing is an interdisciplinary area that deals with the issue of software that is capable of recognising, analysing and reacting to human emotions. The central idea of affective computing is to integrate emotional understanding into intelligent technology systems, usually to improve both the performance and the learning ability of these technologies, for example, by assisting in decisionmaking. The goal is to develop systems that engage with the user in a more natural, adaptive and personalised way. Current affective computing considers ways to combine the methodologies of computer vision, speech research, natural language processing, and machine learning to derive emotional data from facial cues, voice tonal features, physiological metrics, or text conversations [11]. Such advances have led to the applications of emotion-aware systems in the healthcare, educational, customer service, social robotics and human-computer interaction areas.

The theories of emotional intelligence also focus on the importance of being able to perceive, learn, manage and effectively react to emotions in social encounters. These principles applied to AI

systems allow providing more personalised and contextual responses, as well as enhance user satisfaction and interaction. Other recent developments in affective computing have involved multimodal information (i.e. facial expression), speech patterns and textual content used increasingly together to gain a better understanding of emotional states. Despite the many gains realised, the accurate identification of complex, mixed and culturally diversified emotions has been a major issue of research and there has to be further advancements to the theoretical framework and methods of computation [47].

2.1.3 Unimodal Emotion Recognition Systems

As explained by Li and Xu (2025), Unimodal emotion recognition systems use a single modality of information, such as face, voice or text, to identify human emotions [12]. These methods form the basis of emotion recognition research as they are relatively easy to implement and are not as compute-intensive as a multi-modal approach is. With the march of Artificial Intelligence (AI) and Machine Learning, unimodal models have had a remarkable boost in performance and can be used in multiple applications such as facial analysis, voice assistants, sentiment analysis, Healthcare monitoring, etc.

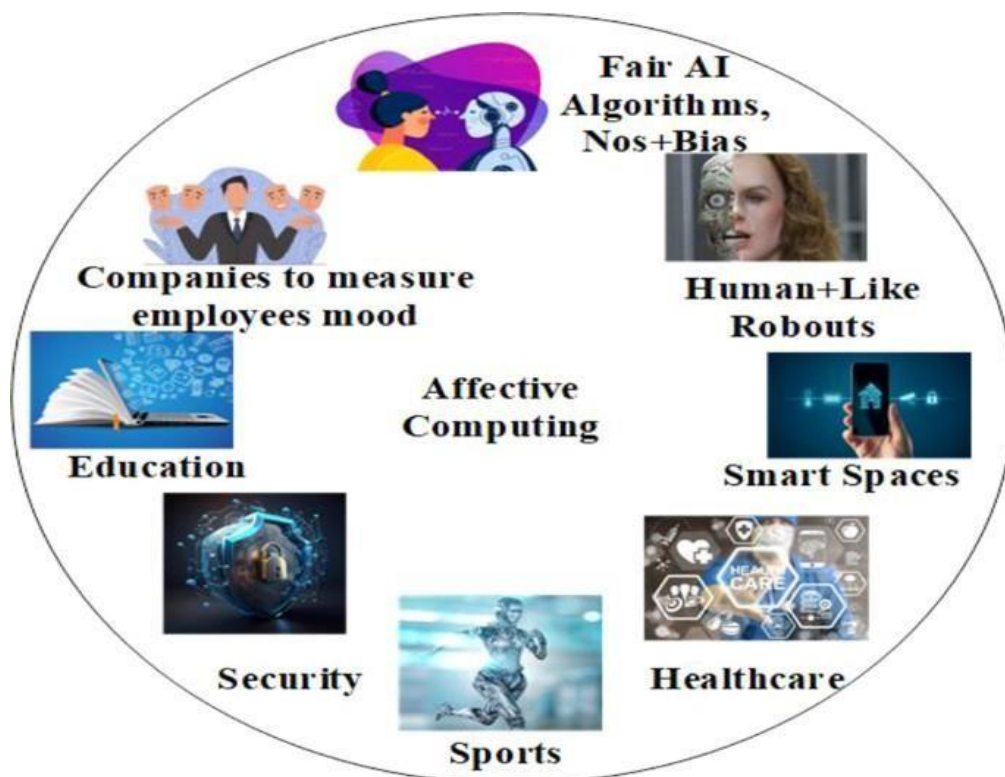


Figure 2.1.3.1: Unimodal Emotion Recognition Systems

(Source: Alsaeedi *et al.* 2025), [13]

Facial Emotion Recognition (FER) is one of the widely explored Unimodal approaches since emotions are reflected through the facial expressions of an individual through rich visual information. Emotions are categorised by analysing various facial features, including eye movement, eyebrow position, mouth shape, and facial muscle movements through the use of AI models. The use of deep learning models has substantially advanced the field of facial emotion recognition, which extracts hierarchical visual features automatically from facial images. Over the last few years, the application of deep learning models has significantly contributed to the recognition of emotions, specifically Convolutional Neural Networks (CNNs), which extract the hierarchical visual characteristics automatically from facial images [14]. Vocal qualities (pitch, tone, tempo/pace, volume, pauses) are affected by human emotion.

Despite the promising performance of the unimodal systems, they tend to depend on environmental factors and the quality of the data. To illustrate, facial recognition systems can be vulnerable to low-light situations, facial coverings or changes in head positioning, whereas speechbased solutions are vulnerable to environmental noise, audio recording, and speaker variability. On the same note, emotion recognition based on text might not be able to read into sarcasm, ambiguity and context. Such constraints diminish the effectiveness of unimodal methods in practical contexts and have deterred researchers from exploring multimodal emotion recognition systems that synthesise complementary information between different sources to realise greater accuracy and strength [48].

2.1.4 Research Gap

Previous studies have shown that AI technology, AIML and AIDL have made tremendous advancements in emotion recognition using machine learning (ML), deep learning (DL), and multimodal data fusion approaches. There are, however, several key research questions that still haven't been answered. A number of the current models are still assessed with only one modality, e.g. facial expression or speech/text, and have not been tested for complex emotional behaviours in real-world scenarios. The comparison of performances for multimodal studies is difficult even within the context of multimodal studies due to the variety of feature extraction methods and level of fusion adopted and to differences in the benchmark data sets.

2.2 Technology Review

2.2.1 Conventional and Deep Learning Approaches for Emotion Recognition

As highlighted by Arjunan (2024), the evolution of emotion recognition systems has progressed from traditional machine learning algorithms to the more sophisticated deep learning architectures [15]. Traditionally, feature extraction has been done by hand, where domain experts determine what features to use in facial images, speech signals, or text, to classify the data. However, deep learning models have the added advantage of being able to learn the hierarchical feature representations on their own from the raw data, leading to better recognition accuracy and generalisations.

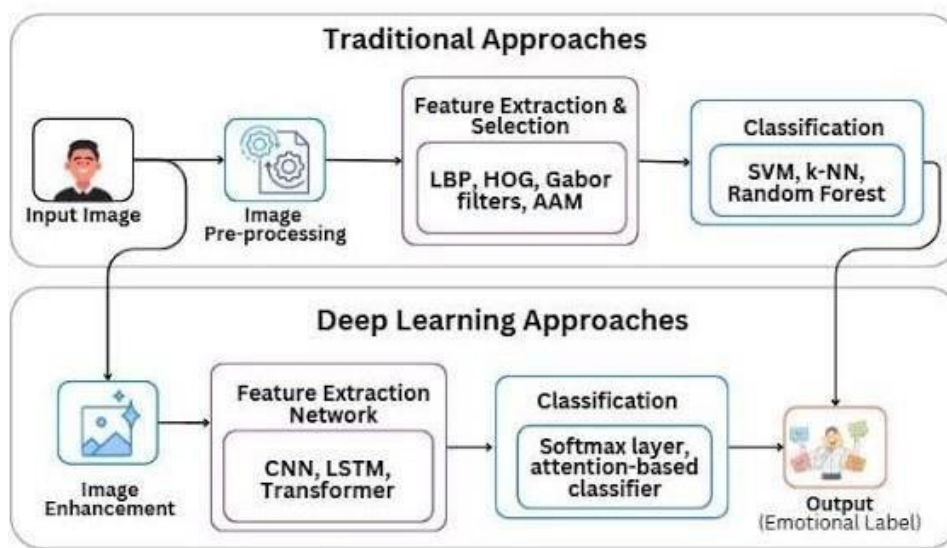


Figure 2.2.1.1: Conventional and Deep Learning Approaches for Emotion Recognition

(Source: Malk & Diwan, 2025) [16]

Early emotion recognition studies have used a variety of traditional machine learning methods. When using manually extracted features, algorithms like Support Vector Machines (SVM), Decision Trees, Random Forests, k-Nearest Neighbour (k-NN) and Naïve Bayes have been successful with certain acceptable results. Various handcrafted features that show the patterns of the face's features have been commonly used for emotion recognition from facial images, like the Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gabor filters [17]. For SED, the primary inputs that have been used are Mel-Frequency Cepstral Coefficients (MFCCs), Pitch, energy, and Spectral features, and in TEER, the extensively used approaches are Text-based features such as Bag-of-Words (BoW), Term Frequency–Inverse Document Frequency (TF-IDF) and Word embedding approaches.

Recent developments in deep learning also added transformer based architectures and selfsupervised learning methods which further enhance emotion recognition performance by acquiring more robust feature representations through large-scale training. This contrasts with conventional machine learning models, which have to contend with manually tuning feature extraction and classification independently, and thus less human intervention and increased scalability [49]. Moreover, developments in Graphics Processing Unit (GPUs) and in cloud computing have made both model training and deployment much faster. Although these advances are made, deep learning models typically need more data with annotations and more computer resources than more traditional approaches, and the consideration of data availability and cost of computer impact should be considered when designing viable emotion recognition systems.

2.2.2 Facial Expression Analysis for Automated Human Emotion Detection

As stated by Shafaqat (2025), Facial Expression Analysis (FEA) is among the most widely employed methods for automatic human emotion detection since facial movements offer rich visual cues of human emotions [18]. Variations in facial muscles – around the eyes, at the eyebrows, on the nose and at the mouth- are used to show happiness, sadness, anger, fear, surprise, disgust and neutrality in human expression. Facial expression recognition is an essential aspect of affective computing, in which Artificial Intelligence (AI) systems analyse these facial features to automatically classify emotions.

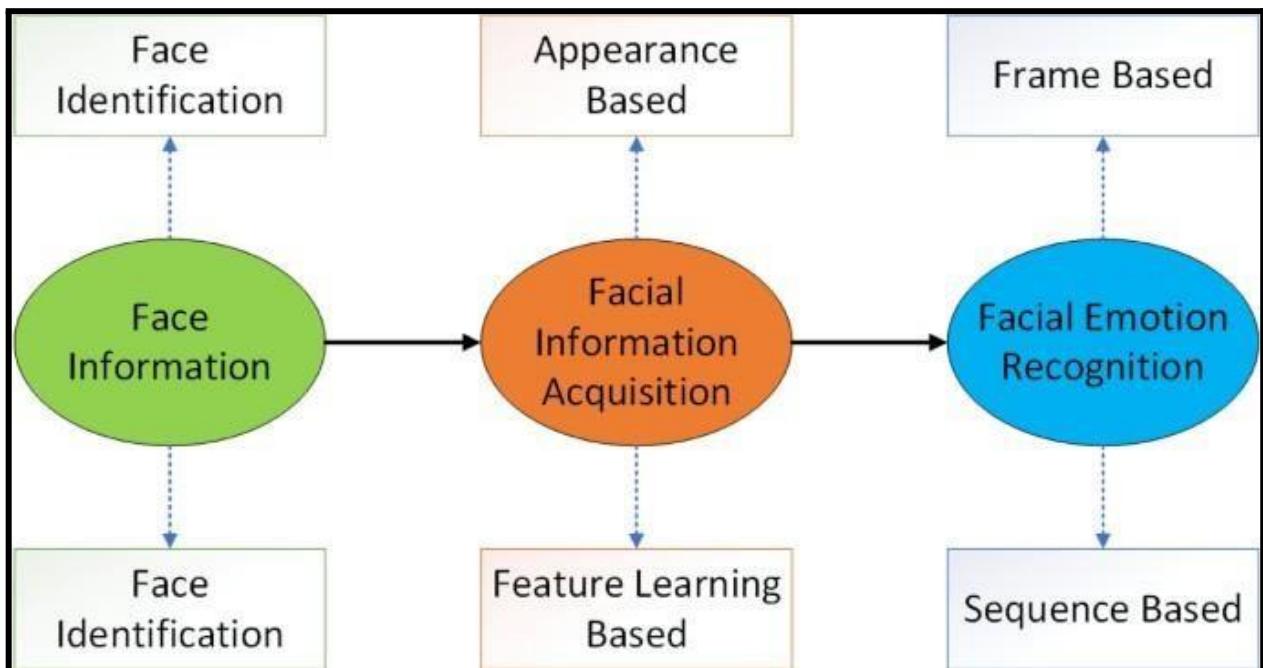


Figure 2.2.2.1: Facial Expression Analysis for Automated Human Emotion Detection

(Source: Mukhtar, 2025) [19]

Owing to the recent advancement in deep learning, facial emotion recognition has been greatly enhanced. Convolutional Neural Networks (CNNs) are capable of automatically learning hierarchical visual features from facial images in the absence of manual feature engineering. CNN-based models can capture slight facial expressions of various moods and improve the classification rate over large datasets. In addition, transfer learning of pre-trained architectures has decreased training time and improved model generalisation in the case of small labelled emotion datasets. Still, there are several problems to be solved. It is hard to classify facial expressions because these are different in different age groups and individuals, and also from one culture to another [20]. When faces are partially masked, worn glasses or hair, or the picture is not in good light, it may be more difficult to recognise the face.

The mechanisms of attention and explainable Artificial Intelligence (XAI) methods have also been considered in recent works and used to enhance the accuracy and transparency of facial emotion recognizers. Tools like Gradient-weighted Class Activation Mapping (Grad-CAM) allow researchers to visualise parts of the face that affect the decisions of models, boosting the trust in AI decision-making. Image rotation, flipping, brightness and scaling are among the data augmentation techniques that are widely used in AI to enhance generalisation of the model and decrease overfitting [50]. Although such developments have been made, it is still hard to realise subtle or mixed expressions of emotion since people tend to exhibit two or more emotions at the same time, which demands more advanced models that can capture finer differences in facial expression.

2.2.3 Multimodal Data Fusion Methods for Emotion Recognition Systems

As reported by Zhang *et al.* (2025), Multimodal data fusion has been considered one of the most important research fields in emotion recognition, because human emotion is discussed as a multiple-modality emotion, in which the sentences are spoken, simultaneously, gestures are displayed, and text representations are used [21]. Multimodal systems combine complementary information from multiple modalities, which makes recognition more accurate and less ambiguous in real-world conditions, as compared with unimodal systems.

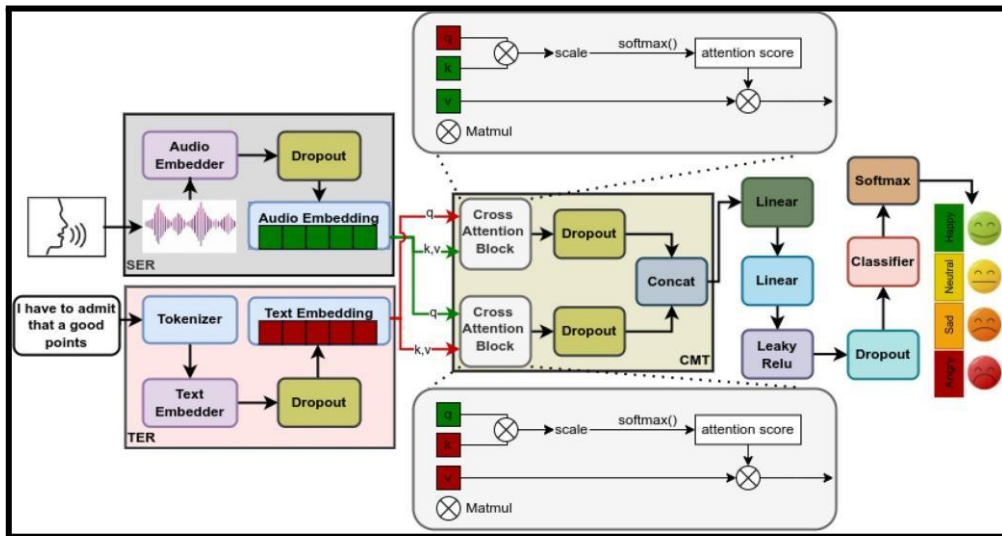


Figure 2.2.3.1: Multimodal Data Fusion Methods for Emotion Recognition Systems

(Source: Chintala, 2024) [22]

There are three main modes of fusion used in the multimodal recognition of emotions. Feature-level fusion is the fusion of extracted features of different modalities into a single modality feature representation before classification. The model can learn correlations between the optical, acoustical and textual features together, which can sometimes lead to better classification accuracy. Feature-level fusion does, however, necessitate feature normalisation since each modality has a different dimensionality. Decision-level fusion is a combination of the prediction outputs from multiple unimodal models. Each classifier produces emotion predictions, and majority voting, weighted averaging, or ensemble learning techniques combine the predictions [23]. This approach is flexible as each modality is used separately, but may miss multi-modality interactions between emotional cues.

The hybrid fusion is another strategy worth considering in the fusion scenarios, as it involves feature-level and decision-level fusion to leverage the advantages of both strategies. Hybrid approaches initially compute complementary representations of features then combine prediction results, commonly leading to better classification performance. The recent multimodal systems are using more and more attention which identifies the relative importance of each modality when predicting emotions, to enable the model to be dynamically adjusted when one of the data sources cannot be relied upon [51]. Despite the importance of multimodal fusion to provide more robustness, issues like modality synchronisation, data loss, complexity in computing, and efficient feature alignment are still part of research to be explored in practical real-time applications.

2.2.4 Deep Learning Architectures for Multimodal Emotion Classification

As discussed by Yoon *et al.* (2024), in the era of deep learning, the automatic extraction of complex emotional features from a variety of data sources has reshaped the landscape of multimodal emotion recognition [24]. Contradistinct from traditional machine learning methods that depend on human-engineered features, deep learning architectures learn features from raw facial images, speech signals and textural information. This functionality has led to much better classification quality and decreased manual effort. Convolutional Neural Networks (CNNs) are still the preferred choice for facial expression analysis because they are able to effectively capture relationships between spatial points in a picture. CNNs automatically recognise the facial features that can be linked to facial expressions such as eye movements, eyebrow positions and mouth shapes. Employing Recurrent Neural Networks (RNNs) to capture sequential dependencies in speech signals has been used to analyse temporal variations in the vocal characteristics effectively for speech analysis. In the case of speech analysis, Recurrent Neural Networks (RNNs) have been commonly used to capture the sequential dependency in speech signals.

More recently, transformer architectures and multimodal attention networks have been shown to achieve promising results, as they successfully model multimodal emotional modalities relationships. The cross-modal attention process permits interaction between facial, speech and textual features during feature learning which enables more emotion rich features to be formed. Transfer learning and foundation models that are pre-trained have also minimised training duration and enhanced generalisation across different datasets. Nonetheless, multimodal deep learning systems are still computationally expensive and can only be effectively trained using large amounts of labelled data. In turn, enhancing computational efficiency and model interpretability and execution on resource-constrained machines is an ongoing or recent field of study.

2.3 Summary

This chapter specifically analysed the previous research studies on Human emotion recognition using Artificial Intelligence. It covered theoretical concepts about affective computing, unimodal emotion recognition systems, traditional machine learning approaches and latest research progress in deep learning. The chapter also explored facial expression analysis, Multimodal Data Fusion Methods and Advanced Deep Learning Architectures for Multimodal Emotion Recognition. Last but not least, the research gaps have been identified as limitations on unimodal approaches, fusion strategies, robustness and applicability in the real world. Based on these findings, a multimodal emotion

recognition system is proposed, and the method followed in developing the system in the next chapter is justified.

Chapter 3 Implementation

3.1 System Design and Architecture

The implementation of the proposed Artificial Intelligence (AI)-based multimodal emotion recognition framework involved the development of deep learning models capable of recognising human emotions from facial images and speech signals. The system has been designed to improve the limitations of unimodal emotion recognition by integrating complementary emotional information obtained from different data modalities. The implementation process consisted of dataset preparation, data preprocessing, development of unimodal deep learning models, multimodal feature fusion, model training and performance evaluation. **The whole framework has** been designed to ensure consistent preprocessing, feature extraction and classification throughout all the experimental conditions.

The first component has been responsible for facial emotion recognition using deep learning models based on images. The second spoke to the speech and used an audio deep learning network to recognize the emotional characteristics from speech recordings. Lastly the extracted multimodal features have been combined into a multimodal framework to perform the final emotion classification. This architecture enabled the framework to learn complementary emotional representations from visual and speech information, thereby improving overall recognition performance.

Preprocessing steps for both benchmark datasets will be undertaken before model training, such as image augmentation, image resizing, audio normalisation and audio feature extraction. Developed models will then be trained, validated and tested with the same experimental conditions, ensuring a fair performance comparison [25]. They will be assessed by classification metrics including accuracy, precision, recall, F1-score and Receiver Operating Characteristic- Area Under the Curve (ROC-AUC). The results of the experiments performed will be analysed to conclude if the performance of emotion recognition can be enhanced after fusing all the available modes of information (multimodal) compared to unimodal deep learning models. Therefore, experimental research methodology will be used as a methodological approach that will make research in the form of experiments be done systematically and objectively to provide a methodology. The validation process of the proposed framework will support the achievement of the research objectives in an objective and reproducible manner.

Selected facial model for fusion: Custom CNN (val_acc=0.4248)		
Facial train embeddings: (24406, 128), Audio train embeddings: (279, 64)		
Fusion train set: (24406, 192), Fusion val set: (4303, 192)		
Model: "Multimodal_Fusion_Model"		
Layer (type)	Output Shape	Param #
=====	=====	=====
fused_embedding_input (InputLayer)	[(None, 192)]	0
dense (Dense)	(None, 128)	24704
batch_normalization_5 (BatchNormalization)	(None, 128)	512
dropout_8 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8256
batch_normalization_6 (BatchNormalization)	(None, 64)	256
dropout_9 (Dropout)	(None, 64)	0
fusion_output (Dense)	(None, 2)	130
=====	=====	=====
Total params: 33,858		
Trainable params: 33,474		
Non-trainable params: 384		

Figure 3.1.1: Multimodal Fusion Model Architecture

Figure 4.2.3.1 shows the Multimodal Fusion Architecture which merges 128-dimensional facial embeddings with 64-dimensional audio embeddings to produce a 192-feature representation [26]. It has 33,858 parameters and achieves two affect classification layers using fully connected Softmax layers successfully in the network.

3.2 Dataset Preparation and Pre-processing

The above pre-processing methods will be carried out on each of the speech and face datasets before processing the model development phase. The image will be subjected to some image processing techniques such as resizing, pixel normalization and image augmentation to improve feature learning in images and generalization of the model. Some of the methods for audio pre-processing will include waveform normalization, removal of silences and feature extraction for normalizing the input data used for speech emotion recognition. During the experimentation process, there will be construction of several deep learning models. The emotional recognition on faces will be carried out based on a Custom CNN model and efficientNetB0 model through transfer learning. A CNN-BiLSTM model for speech emotion recognition will be developed since it combines the two functions of convolution and sequence learning.

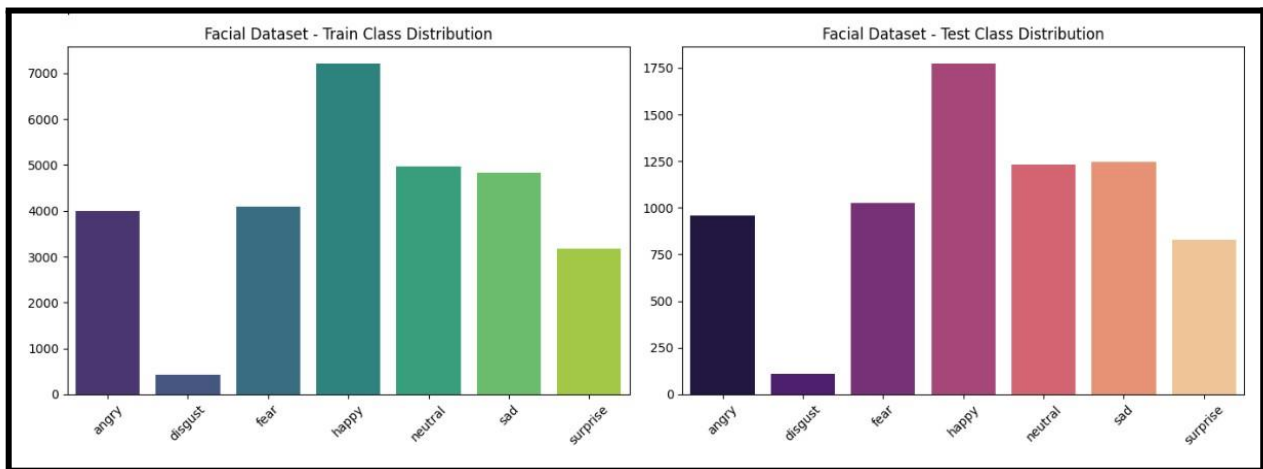


Figure 3.2.1: Facial Testing Dataset Class Distribution

The distribution of the facial testing dataset with 7 emotion classes is shown in Figure 4.2.1.2. Happy captures the highest number of 1774 images, disgust has the lowest number with only 111 images. The other categories vary from 831 to 1247 samples and allow the evaluation of the models with a good performance across all the emotions.

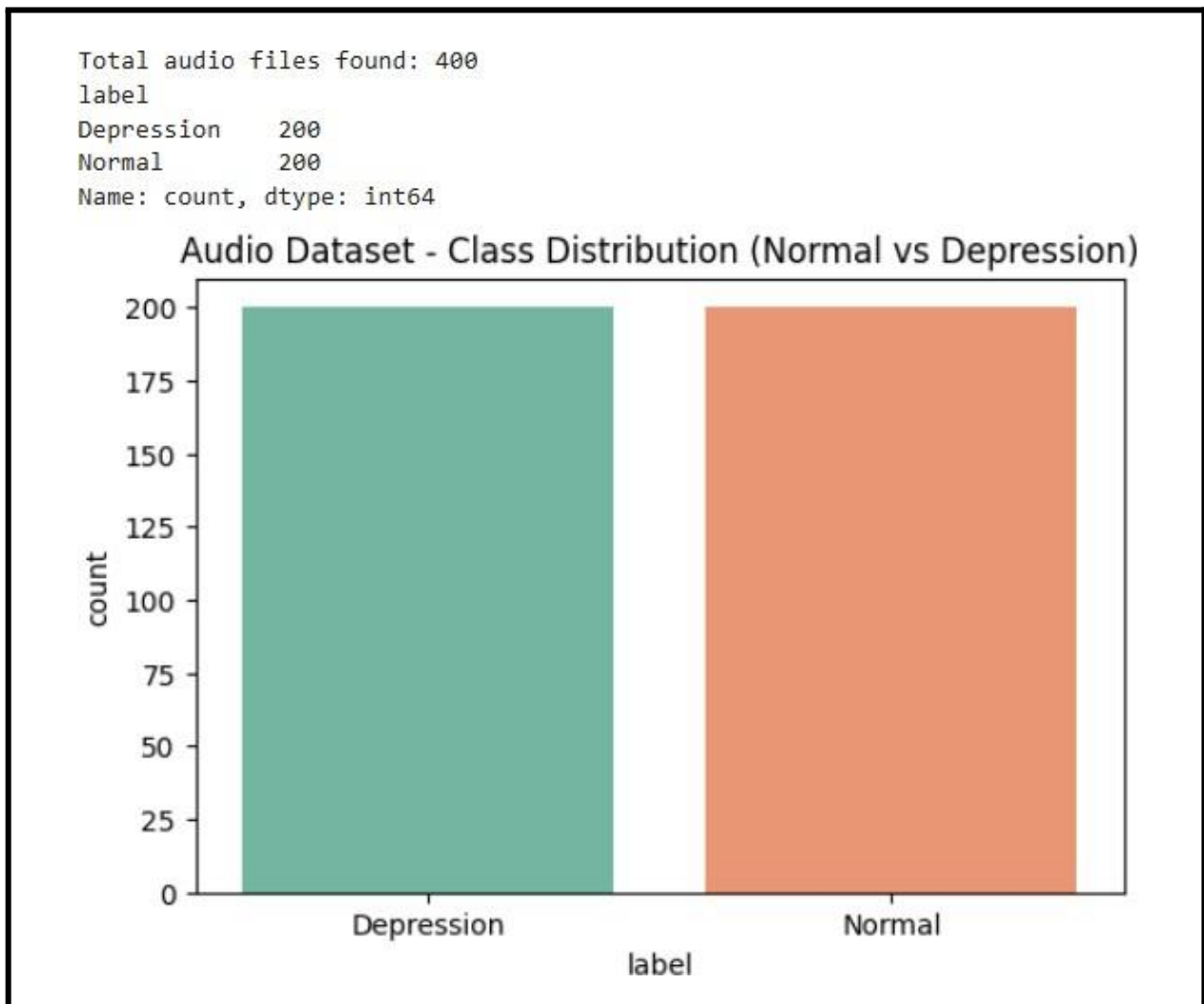


Figure 3.2.2: Audio Dataset Class Distribution

The distribution of audio data used in this study is shown in Figure 4.2.1.3 and is comprised of 400 speech recordings. Each of the two groups (Depression and Normal) has 200 samples, making for a well-balanced binary dataset [27]. This is a balanced distribution to reduce class bias and to guarantee a reliable classification of audio emotions during the training of the model.

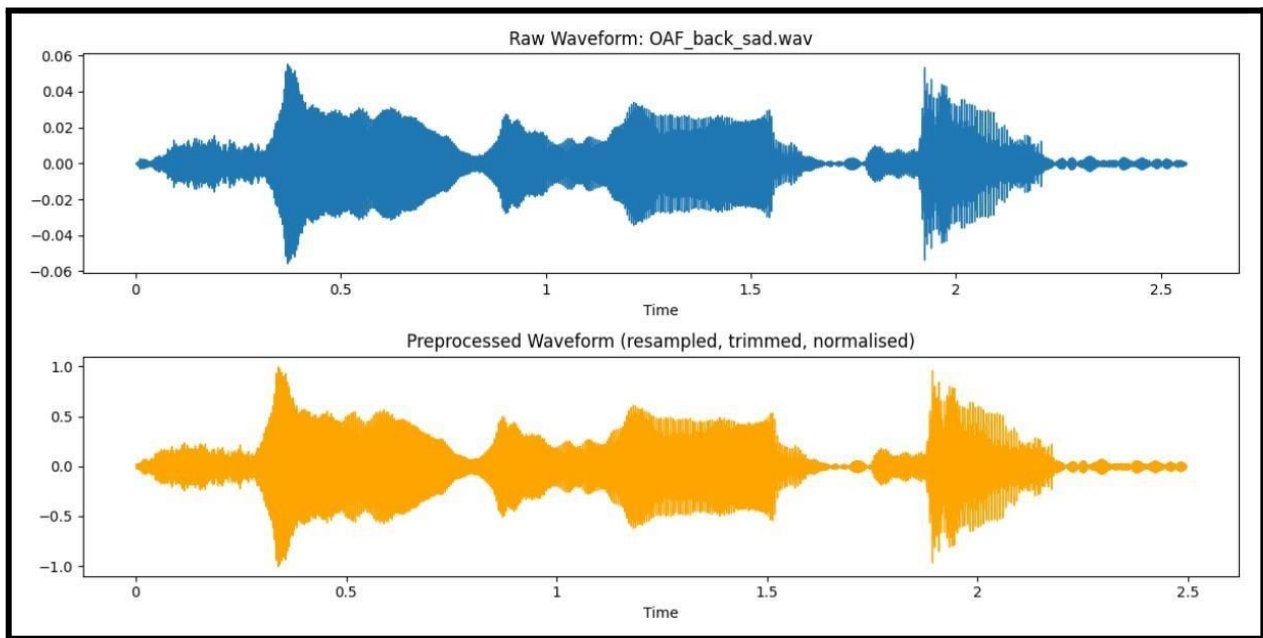


Figure 3.2.3: Raw and Preprocessed Audio Waveforms

Figure 4.2.1.4 shows raw and preprocessed audio speech waveforms of a depression speech sample. Preprocessing rescales the signal, removes silences and normalizes the amplitude of the signal, resulting in a more consistent signal of about the same amplitude. These enhancements improve the quality of the feature extraction and learning efficiency in audio classification.

```
Processed 25/400 audio files...
Processed 50/400 audio files...
Processed 75/400 audio files...
Processed 100/400 audio files...
Processed 125/400 audio files...
Processed 150/400 audio files...
Processed 175/400 audio files...
Processed 200/400 audio files...
Processed 225/400 audio files...
Processed 250/400 audio files...
Processed 275/400 audio files...
Processed 300/400 audio files...
Processed 325/400 audio files...
Processed 350/400 audio files...
Processed 375/400 audio files...
Processed 400/400 audio files...

Feature extraction complete. Extracted 400 feature vectors of dimensionality 192.
Skipped 0 corrupted/unreadable files.

Label mapping: {'Depression': 0, 'Normal': 1}
Feature matrix shape: (400, 192)
Label vector shape : (400,)
```

Figure 3.2.4: Audio Feature Extraction Summary

The summary of audio feature extraction process is summarized by Figure 4.2.1.5. Successfully processed all 400 recordings without corrupt records. The extracted feature matrix consists of 400 samples and 192 numerical features, which can be used to evaluate and train CNN-BiLSTM model with detailed acoustic information.



Figure 3.2.5: Sample Augmented Training Images

Figure 4.2.1.6 shows the augmented facial training images for rotation, zoom, translation and horizontal flipping. Label for emotion is preserved, and the augmentation process makes diversity in images. This would help to improve the generalization of the model and to decrease the overfitting effectively in the training of the facial emotion recognition model [28].

3.3 Development of Unimodal Models

Audio CNN-BiLSTM Model

The quantitative approach will allow for systematically comparing different deep learning architectures, such as Custom Convolutional Neural Network (CNN), EfficientNetB0 transfer learning model, Audio CNN-BiLSTM model, as well as the proposed multimodal fusion approach. All the models will be tested using the same experimental setup to understand the classification power of each model under equal conditions. Lastly, the quantitative research technique will help facilitate the reliability and repeatability of this study, as everyone will use the same preprocessing techniques, model configurations and assessment metrics during the experimental process [29].

Such an approach can serve as a robust and objective measurement to assess the efficacy of multimodal deep learning for automated human emotion detection.

Model: "Audio_CNN_BiLSTM"		
Layer (type)	Output Shape	Param #
=====		
audio_features_input (Input Layer)	[(None, 192, 1)]	0
conv1d (Conv1D)	(None, 192, 64)	256
batch_normalization (Batch Normalization)	(None, 192, 64)	256
max_pooling1d (MaxPooling1D)	(None, 96, 64)	0
dropout (Dropout)	(None, 96, 64)	0
conv1d_1 (Conv1D)	(None, 96, 128)	24704
batch_normalization_1 (Batch Normalization)	(None, 96, 128)	512
max_pooling1d_1 (MaxPooling1D)	(None, 48, 128)	0
dropout_1 (Dropout)	(None, 48, 128)	0
bidirectional (Bidirectional)	(None, 128)	98816
dropout_2 (Dropout)	(None, 128)	0
audio_embedding (Dense)	(None, 64)	8256
dropout_3 (Dropout)	(None, 64)	0
audio_output (Dense)	(None, 2)	130
=====		
Total params: 132,930		
Trainable params: 132,546		
Non-trainable params: 384		
=====		
Audio class weights: {0: 1.0035971223021583, 1: 0.9964285714285714}		

Figure 3.3.1: Audio CNN-BiLSTM Model Architecture

The proposed Audio CNN-BiLSTM network with Conv1D, Batch Normalization, MaxPooling, BiLSTM, and Dense layers is shown in this figure. It successfully classifies the emotions of the input images

by using 132,930 total parameters in the model, which processes 192 input features and has a Softmax output layer with a 2-class output.

```
Epoch 1/10
9/9 [=====] - 0s 46ms/step - loss: 0.2306 - accuracy: 0.9211 - precision: 0.9211 - recall: 0.9211 - auc: 0.9655 - val_loss: 0.5911 - val_accuracy: 0.7000 - val_precision: 0.7000 - val_recall: 0.7000 - val_auc: 0.8406 - lr: 0.0010
Epoch 2/10
9/9 [=====] - 0s 22ms/step - loss: 0.2105 - accuracy: 0.9355 - precision: 0.9355 - recall: 0.9355 - auc: 0.9714 - val_loss: 0.5989 - val_accuracy: 0.7000 - val_precision: 0.7000 - val_recall: 0.7000 - val_auc: 0.8294 - lr: 0.0010
Epoch 3/10
9/9 [=====] - 0s 19ms/step - loss: 0.2041 - accuracy: 0.9283 - precision: 0.9283 - recall: 0.9283 - auc: 0.9717 - val_loss: 0.6006 - val_accuracy: 0.6750 - val_precision: 0.6750 - val_recall: 0.6750 - val_auc: 0.8256 - lr: 0.0010
Epoch 4/10
9/9 [=====] - 0s 19ms/step - loss: 0.1468 - accuracy: 0.9498 - precision: 0.9498 - recall: 0.9498 - auc: 0.9895 - val_loss: 0.5951 - val_accuracy: 0.7000 - val_precision: 0.7000 - val_recall: 0.7000 - val_auc: 0.8331 - lr: 0.0010
Epoch 5/10
9/9 [=====] - 0s 28ms/step - loss: 0.1193 - accuracy: 0.9606 - precision: 0.9606 - recall: 0.9606 - auc: 0.9924 - val_loss: 0.5886 - val_accuracy: 0.6500 - val_precision: 0.6500 - val_recall: 0.6500 - val_auc: 0.8234 - lr: 0.0010
Epoch 6/10
9/9 [=====] - 0s 19ms/step - loss: 0.0928 - accuracy: 0.9749 - precision: 0.9749 - recall: 0.9749 - auc: 0.9960 - val_loss: 0.5896 - val_accuracy: 0.6500 - val_precision: 0.6500 - val_recall: 0.6500 - val_auc: 0.8250 - lr: 0.0010
Epoch 7/10
9/9 [=====] - 0s 27ms/step - loss: 0.1087 - accuracy: 0.9713 - precision: 0.9713 - recall: 0.9713 - auc: 0.9914 - val_loss: 0.5834 - val_accuracy: 0.6750 - val_precision: 0.6750 - val_recall: 0.6750 - val_auc: 0.8362 - lr: 0.0010
Epoch 8/10
9/9 [=====] - 0s 25ms/step - loss: 0.0702 - accuracy: 0.9821 - precision: 0.9821 - recall: 0.9821 - auc: 0.9974 - val_loss: 0.5692 - val_accuracy: 0.7000 - val_precision: 0.7000 - val_recall: 0.7000 - val_auc: 0.8575 - lr: 0.0010
Epoch 9/10
9/9 [=====] - 0s 25ms/step - loss: 0.0497 - accuracy: 0.9857 - precision: 0.9857 - recall: 0.9857 - auc: 0.9988 - val_loss: 0.5632 - val_accuracy: 0.7000 - val_precision: 0.7000 - val_recall: 0.7000 - val_auc: 0.8513 - lr: 0.0010
Epoch 10/10
9/9 [=====] - 0s 29ms/step - loss: 0.0340 - accuracy: 0.9892 - precision: 0.9892 - recall: 0.9892 - auc: 0.9997 - val_loss: 0.5612 - val_accuracy: 0.6750 - val_precision: 0.6750 - val_recall: 0.6750 - val_auc: 0.8306 - lr: 0.0010
```

Figure 3.3.2: Audio CNN-BiLSTM Training Progress

Figure 4.2.2.2 presents the Audio CNN-BiLSTM training progress across ten epochs. The accuracy rate for the training is raised from 92.11% to 98.92% while the accuracy rate of the validation fluctuates between 65% and 70%. The learning process shows consistent optimization, that is, the values of the learning loss are continuously decreasing.

3.4 Custom CNN Model

Model: "Custom_CNN"		
Layer (type)	Output Shape	Param #
facial_image_input (InputLayer)	[(None, 224, 224, 3)]	0
conv2d (Conv2D)	(None, 224, 224, 32)	896
batch_normalization_2 (BatchNormalization)	(None, 224, 224, 32)	128
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
dropout_4 (Dropout)	(None, 112, 112, 32)	0
conv2d_1 (Conv2D)	(None, 112, 112, 64)	18496
batch_normalization_3 (BatchNormalization)	(None, 112, 112, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
dropout_5 (Dropout)	(None, 56, 56, 64)	0
conv2d_2 (Conv2D)	(None, 56, 56, 128)	73856
batch_normalization_4 (BatchNormalization)	(None, 56, 56, 128)	512
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0
custom_cnn_embedding (Dense)	(None, 128)	16512
dropout_6 (Dropout)	(None, 128)	0
custom_cnn_output (Dense)	(None, 7)	903
=====		
Total params: 111,559		
Trainable params: 111,111		
Non-trainable params: 448		

Figure 3.4.1: Custom CNN Model Architecture

The architecture of the Custom CNN is shown in Figure 4.2.2.5 with three convolutional blocks, Batch Normalization, Global Average Pooling and Dense layers. This network is able to classify seven facial emotion classes with success on facial images of 224×224 pixels and 111,559 parameters.

```
Epoch 1/10
763/763 [=====] - 214s 280ms/step - loss: 1.6855 - accuracy: 0.3230 - precision: 0.6010 - recall: 0.0335 - auc: 0.7261 - val_loss: 6.7428 - val_accuracy: 0.1418 - val_precision: 0.1390 - val_recall: 0.1246 - val_auc: 0.5127 - lr: 0.0010
Epoch 2/10
763/763 [=====] - 211s 276ms/step - loss: 1.6601 - accuracy: 0.3420 - precision: 0.6178 - recall: 0.0515 - auc: 0.7381 - val_loss: 1.9205 - val_accuracy: 0.2803 - val_precision: 0.3246 - val_recall: 0.2045 - val_auc: 0.7015 - lr: 0.0010
Epoch 3/10
763/763 [=====] - 212s 277ms/step - loss: 1.6357 - accuracy: 0.3538 - precision: 0.6323 - recall: 0.0656 - auc: 0.7488 - val_loss: 1.7529 - val_accuracy: 0.2705 - val_precision: 0.7778 - val_recall: 0.0016 - val_auc: 0.6887 - lr: 0.0010
Epoch 4/10
763/763 [=====] - 220s 288ms/step - loss: 1.6078 - accuracy: 0.3675 - precision: 0.6458 - recall: 0.0853 - auc: 0.7596 - val_loss: 2.7334 - val_accuracy: 0.2603 - val_precision: 0.2684 - val_recall: 0.2403 - val_auc: 0.6735 - lr: 0.0010
Epoch 5/10
763/763 [=====] - 218s 285ms/step - loss: 1.5767 - accuracy: 0.3864 - precision: 0.6616 - recall: 0.1094 - auc: 0.7708 - val_loss: 3.0218 - val_accuracy: 0.2726 - val_precision: 0.2779 - val_recall: 0.2240 - val_auc: 0.6502 - lr: 0.0010
Epoch 6/10
763/763 [=====] - 214s 280ms/step - loss: 1.5501 - accuracy: 0.3939 - precision: 0.6759 - recall: 0.1268 - auc: 0.7796 - val_loss: 1.9445 - val_accuracy: 0.2858 - val_precision: 0.3791 - val_recall: 0.0860 - val_auc: 0.6899 - lr: 0.0010
Epoch 7/10
763/763 [=====] - 219s 287ms/step - loss: 1.5336 - accuracy: 0.4029 - precision: 0.6853 - recall: 0.1396 - auc: 0.7854 - val_loss: 2.1452 - val_accuracy: 0.2766 - val_precision: 0.3735 - val_recall: 0.1195 - val_auc: 0.6700 - lr: 0.0010
Epoch 8/10
763/763 [=====] - 211s 276ms/step - loss: 1.5021 - accuracy: 0.4151 - precision: 0.7003 - recall: 0.1562 - auc: 0.7955 - val_loss: 1.4881 - val_accuracy: 0.4209 - val_precision: 0.6790 - val_recall: 0.1922 - val_auc: 0.7997 - lr: 5.0000e-04
Epoch 9/10
763/763 [=====] - 209s 274ms/step - loss: 1.4866 - accuracy: 0.4244 - precision: 0.6994 - recall: 0.1635 - auc: 0.8014 - val_loss: 1.5634 - val_accuracy: 0.3944 - val_precision: 0.6429 - val_recall: 0.1569 - val_auc: 0.7785 - lr: 5.0000e-04
Epoch 10/10
763/763 [=====] - 211s 276ms/step - loss: 1.4767 - accuracy: 0.4305 - precision: 0.7024 - recall: 0.1720 - auc: 0.8046 - val_loss: 1.4984 - val_accuracy: 0.4248 - val_precision: 0.7147 - val_recall: 0.1648 - val_auc: 0.7944 - lr: 5.0000e-04
```

Figure 3.4.2: Custom CNN Training Progress

The training process of the Custom CNN is shown in figure 4.2.2.6 over 10 epochs. The accuracy of the training set increases from 32.30% to 43.05% and the validation set increases from 14.18%. The training loss is found to reduce gradually, having shown gradual learning of the features even with the difficult multi-class facial recognition problem overall.

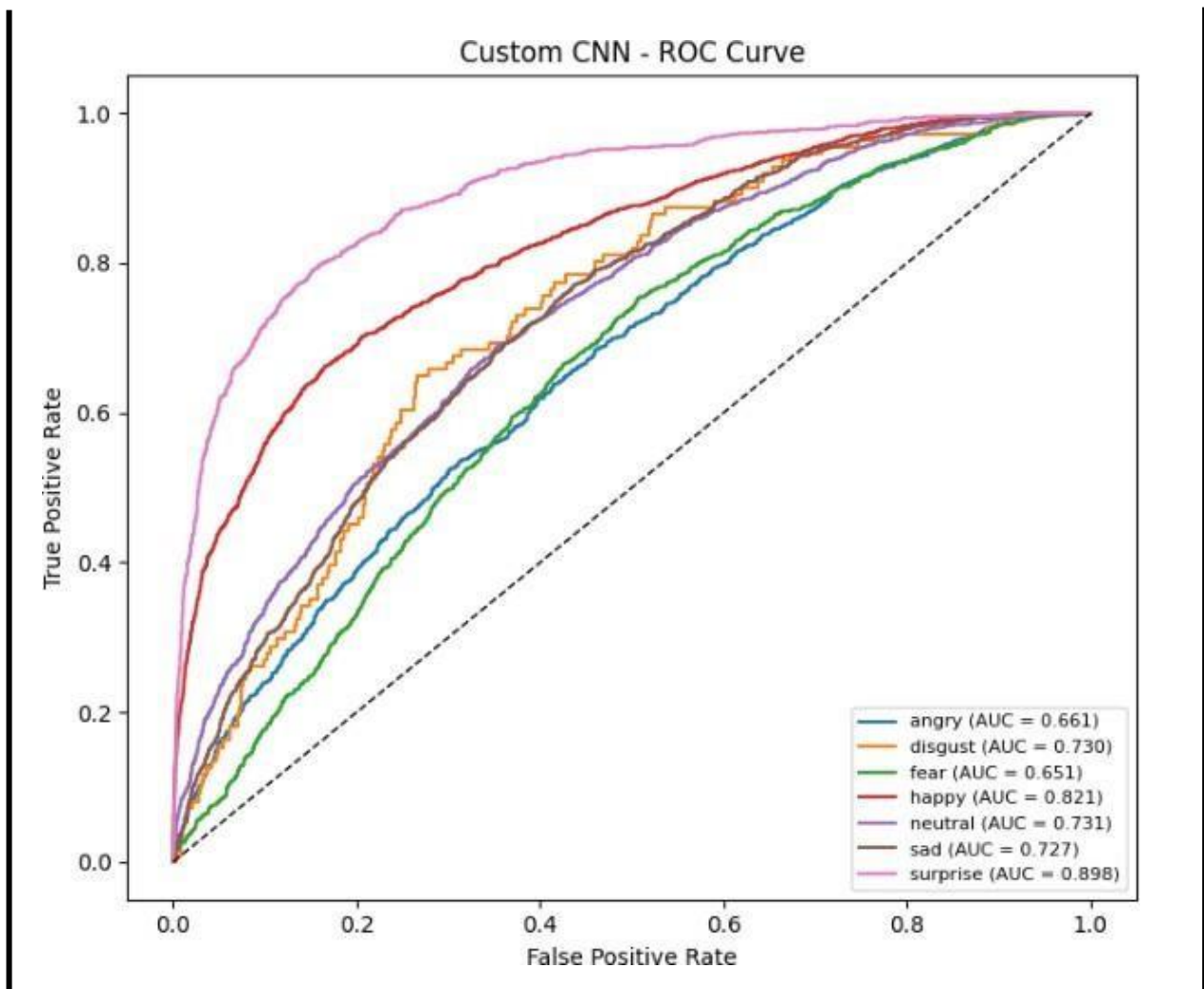


Figure 3.4.3: Custom CNN ROC Curve

The ROC curves of seven facial emotion classes are shown in Figure 4.2.2.8. Surprise has the highest AUC score of 0.898, while happy has an AUC of 0.821. The smallest AUC is found for Fear (0.651), reflecting inter-categorical differences in discrimination.

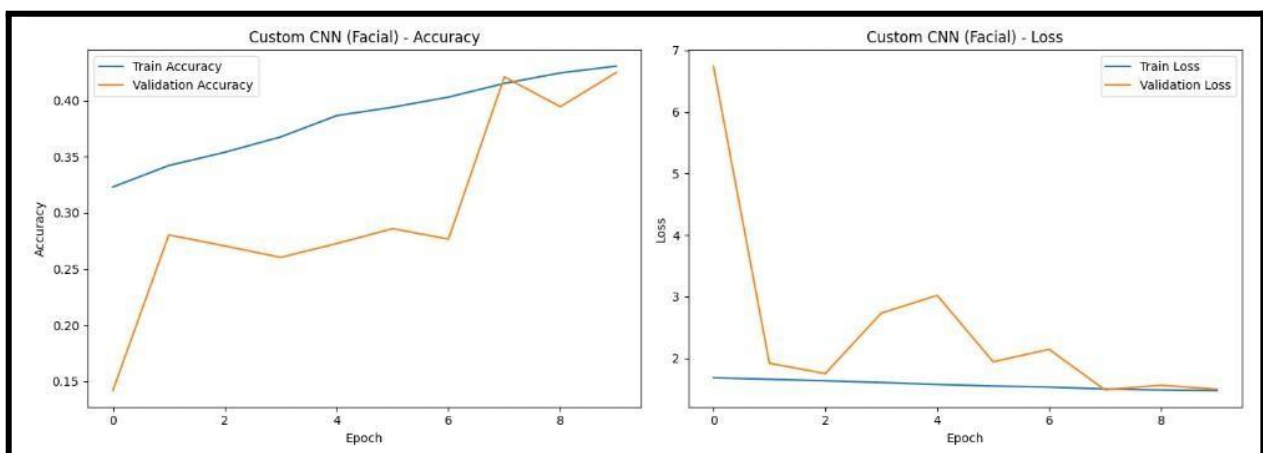


Figure 3.4.4: Custom CNN Training Curves

The Custom CNN accuracy and loss curves are shown in Figure 4.2.2.9. The accuracy for training gradually rises while the accuracy for validation varies and settles around 42.48%. The training loss is reduced steadily over the training epochs and the validation loss does not move away from the acceptable value over the course of the optimization process successfully.

3.5 EfficientNetB0 Transfer Learning Model

Model: "EfficientNetB0_Transfer"		
Layer (type)	Output Shape	Param #
=====		
facial_image_input_effnet (InputLayer)	[(None, 224, 224, 3)]	0
efficientnetb0 (Functional)	(None, 7, 7, 1280)	4049571
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1280)	0
effnet_embedding (Dense)	(None, 128)	163968
dropout_7 (Dropout)	(None, 128)	0
effnet_output (Dense)	(None, 7)	903
=====		
Total params: 4,214,442		
Trainable params: 164,871		
Non-trainable params: 4,049,571		

Figure 3.5.1: EfficientNetB0 Model Architecture

The EfficientNetB0 transfer learning architecture is shown in figure 4.2.2.10. Before Dense classification layers, the pretrained ImageNet backbone detects facial features. The network has 4,214,442 parameters, among which 164,871 parameters are trainable, for efficient learning of the features for 7 facial emotion classes successfully.

```

Epoch 1/10
763/763 [=====] - 220s 283ms/step - loss: 1.8311 - accuracy: 0.2407 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6344
- val_loss: 1.8108 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6425 - lr: 0.0010
Epoch 2/10
763/763 [=====] - 217s 284ms/step - loss: 1.8189 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6411
- val_loss: 1.8102 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6436 - lr: 0.0010
Epoch 3/10
763/763 [=====] - 213s 279ms/step - loss: 1.8164 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6431
- val_loss: 1.8106 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6433 - lr: 0.0010
Epoch 4/10
763/763 [=====] - 215s 281ms/step - loss: 1.8167 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6429
- val_loss: 1.8096 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 0.0010
Epoch 5/10
763/763 [=====] - 214s 280ms/step - loss: 1.8151 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6434
- val_loss: 1.8102 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6431 - lr: 0.0010
Epoch 6/10
763/763 [=====] - 214s 281ms/step - loss: 1.8145 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6427
- val_loss: 1.8101 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6431 - lr: 0.0010
Epoch 7/10
763/763 [=====] - 216s 284ms/step - loss: 1.8135 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6424
- val_loss: 1.8097 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6436 - lr: 0.0010
Epoch 8/10
763/763 [=====] - 212s 278ms/step - loss: 1.8116 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6430
- val_loss: 1.8108 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6435 - lr: 5.0000e-04
Epoch 9/10
763/763 [=====] - 214s 280ms/step - loss: 1.8113 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6433
- val_loss: 1.8094 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 5.0000e-04
Epoch 10/10
763/763 [=====] - 208s 273ms/step - loss: 1.8114 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6428
- val_loss: 1.8106 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 5.0000e-04

```

Figure 3.5.2: EfficientNetB0 Phase 1 Training Progress

Figure 4.2.2.11 presents Phase One training using the frozen EfficientNetB0 backbone. In the absence of fine-tuning the pretrained feature extraction layers during optimization, training and validation accuracy of about 25.13% are seen for 10 epochs.

```

Epoch 1/10
763/763 [=====] - 217s 279ms/step - loss: 1.8110 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6433
- val_loss: 1.8100 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 1.0000e-05
Epoch 2/10
763/763 [=====] - 208s 273ms/step - loss: 1.8110 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6437
- val_loss: 1.8095 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 1.0000e-05
Epoch 3/10
763/763 [=====] - 216s 283ms/step - loss: 1.8109 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6435
- val_loss: 1.8098 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 1.0000e-05
Epoch 4/10
763/763 [=====] - 218s 286ms/step - loss: 1.8103 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6436
- val_loss: 1.8094 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 1.0000e-05
Epoch 5/10
763/763 [=====] - 215s 281ms/step - loss: 1.8111 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6432
- val_loss: 1.8097 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 1.0000e-05
Epoch 6/10
763/763 [=====] - 211s 277ms/step - loss: 1.8110 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6437
- val_loss: 1.8097 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 5.0000e-06
Epoch 7/10
763/763 [=====] - 222s 291ms/step - loss: 1.8112 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6435
- val_loss: 1.8099 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 5.0000e-06
Epoch 8/10
763/763 [=====] - 214s 281ms/step - loss: 1.8102 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6435
- val_loss: 1.8096 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 5.0000e-06
Epoch 9/10
763/763 [=====] - 211s 277ms/step - loss: 1.8104 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6437
- val_loss: 1.8096 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6439 - lr: 2.5000e-06
Epoch 10/10
763/763 [=====] - 220s 288ms/step - loss: 1.8108 - accuracy: 0.2513 - precision: 0.0000e+00 - recall: 0.0000e+00 - auc: 0.6440
- val_loss: 1.8096 - val_accuracy: 0.2515 - val_precision: 0.0000e+00 - val_recall: 0.0000e+00 - val_auc: 0.6437 - lr: 2.5000e-06

```

Figure 3.5.3: EfficientNetB0 Phase 2 Fine-Tuning Progress

The results for Fine-Tuning with partially unfrozen EfficientNetB0 layers are shown in Figure 4.2.2.12 for Phase Two. The accuracy of training and validation are still around 25.13%, and the loss varies

slightly. Fine-tuning gives only small performance boosts, indicating that it is not effective for altering the adaptation of the selected facial emotion data set.

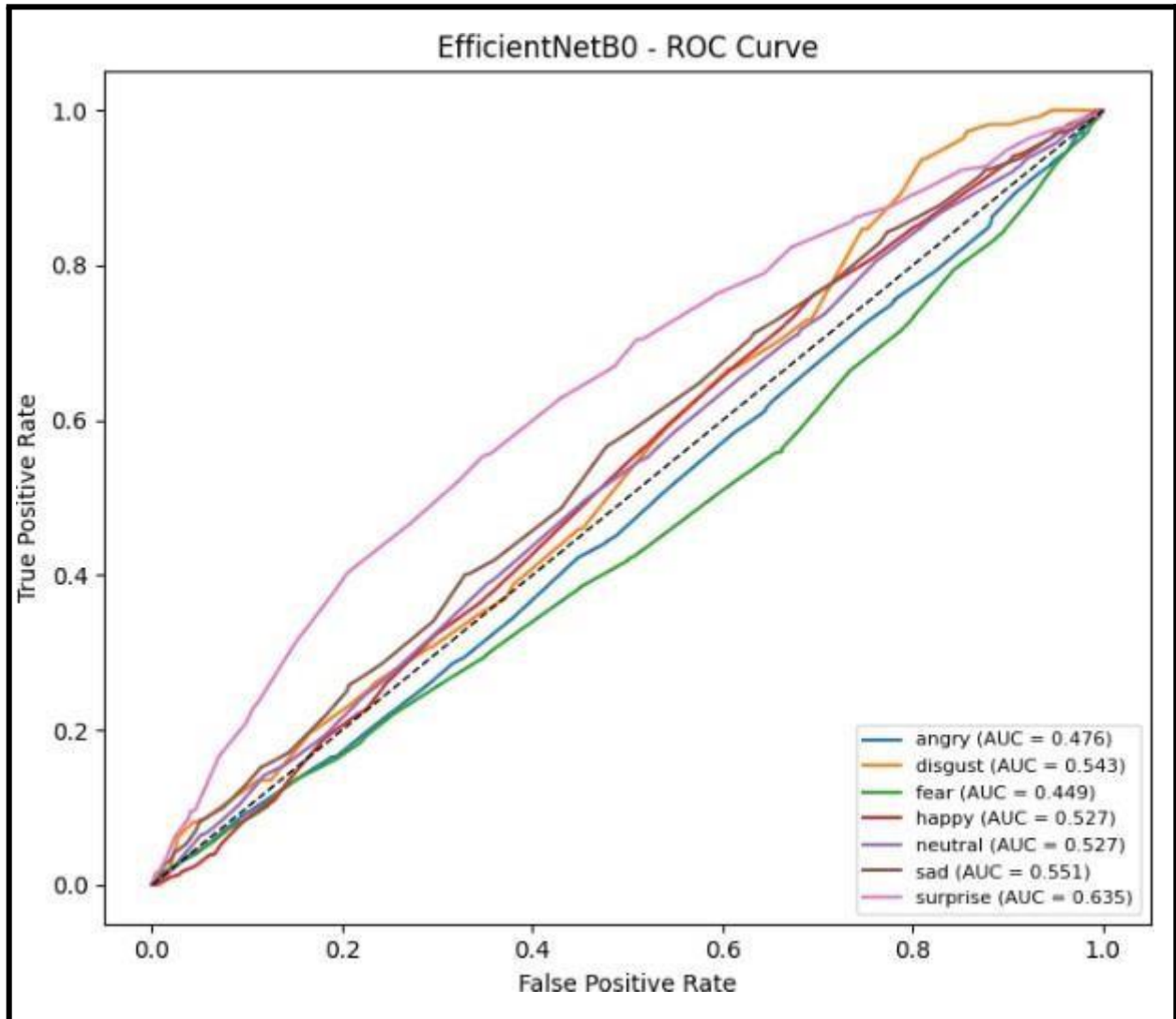


Figure 3.5.4: EfficientNetB0 ROC Curve

In figure 4.2.2.14, the EfficientNetB0 ROC curves for each of the seven emotion classes are shown. Surprise has the largest AUC (0.635) and fear has the smallest AUC (0.449). The discrimination performance is still low, overall, in all facial emotion categories for evaluation.

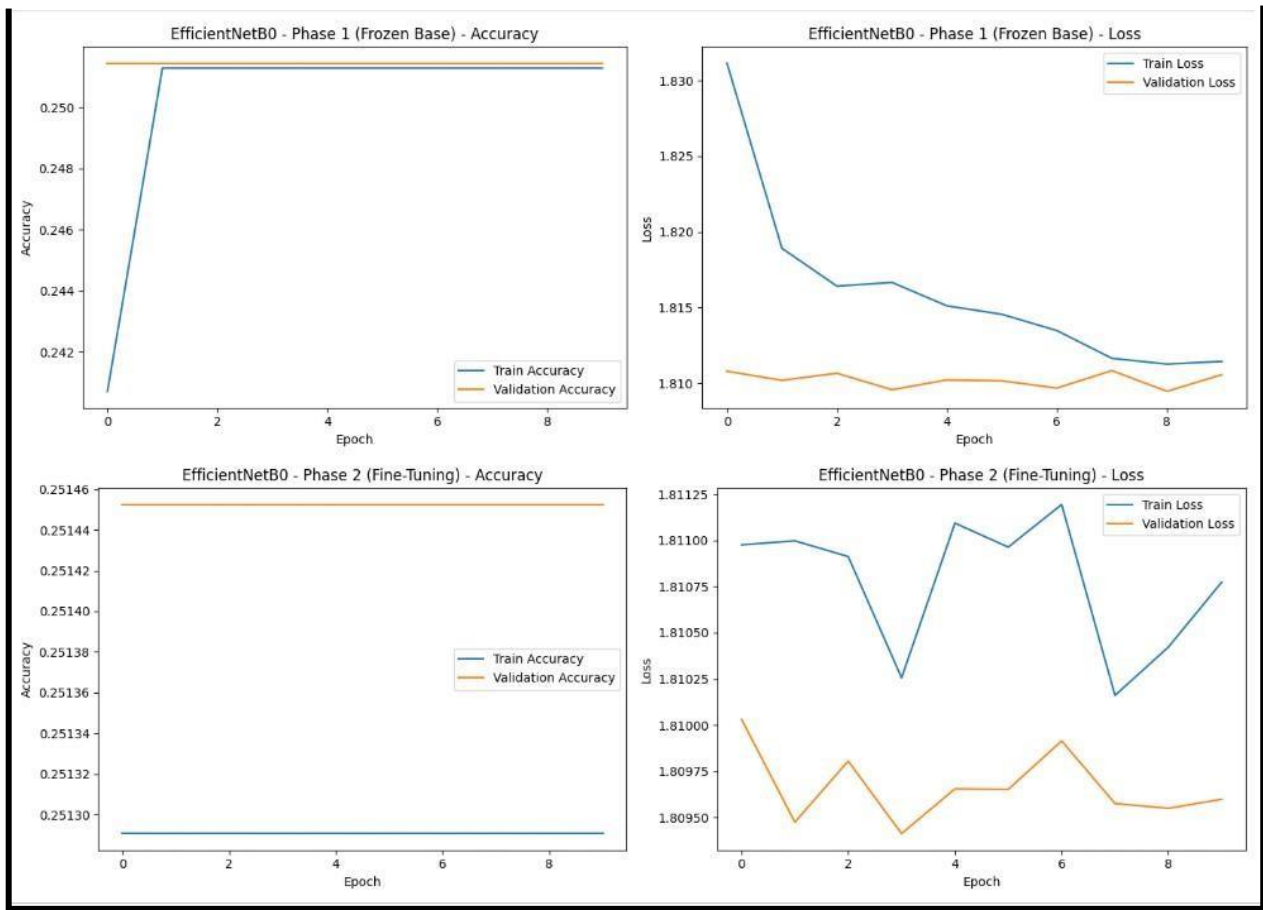


Figure 3.5.5: EfficientNetB0 Training Curves

The EfficientNetB0 training and validation curves for both learning phases are shown in Figure 4.2.2.15. The accuracy remains almost constant at 25.13%, and loss shows almost negligible variations. The observations show that the transfer learning and fine-tuning process did not lead to significant learning.

3.6 Multimodal Fusion Framework

The developed deep learning models will be evaluated in terms of quantitative methods. Transition to a commonly accepted classification metric – accuracy, precision, recall, F1-score and ROC-AUC [30]. Each of them can be assessed in terms of these performance indicators, which will give an objective measurement of the effectiveness with which each model recognises human emotions. Confusion matrices will also be discussed to measure the performance of the prediction of the emotion class-wise, and to find out the emotions classified correctly and the misclassified ones. Discussions on learning curves to see how well the model is trained or validated and identify any signs of overfitting or underfitting those models will also be performed. Progressively, ROC curves

will be used for assessing the discrimination power of each emotion class. The unimodal models will be compared with the proposed multimodal fusion framework.

```
Epoch 1/10
763/763 [=====] - 6s 7ms/step - loss: 0.1686 - accuracy: 0.9343 - precision: 0.9343 - recall: 0.9343 - auc: 0.9827 - val_loss:
0.0832 - val_accuracy: 0.9677 - val_precision: 0.9677 - val_recall: 0.9677 - val_auc: 0.9967 - lr: 0.0010
Epoch 2/10
763/763 [=====] - 5s 6ms/step - loss: 0.0845 - accuracy: 0.9685 - precision: 0.9685 - recall: 0.9685 - auc: 0.9952 - val_loss:
0.0641 - val_accuracy: 0.9733 - val_precision: 0.9733 - val_recall: 0.9733 - val_auc: 0.9977 - lr: 0.0010
Epoch 3/10
763/763 [=====] - 5s 6ms/step - loss: 0.0656 - accuracy: 0.9753 - precision: 0.9753 - recall: 0.9753 - auc: 0.9971 - val_loss:
0.0509 - val_accuracy: 0.9798 - val_precision: 0.9798 - val_recall: 0.9798 - val_auc: 0.9987 - lr: 0.0010
Epoch 4/10
763/763 [=====] - 5s 7ms/step - loss: 0.0612 - accuracy: 0.9771 - precision: 0.9771 - recall: 0.9771 - auc: 0.9974 - val_loss:
0.0738 - val_accuracy: 0.9689 - val_precision: 0.9689 - val_recall: 0.9689 - val_auc: 0.9968 - lr: 0.0010
Epoch 5/10
763/763 [=====] - 5s 7ms/step - loss: 0.0554 - accuracy: 0.9788 - precision: 0.9788 - recall: 0.9788 - auc: 0.9980 - val_loss:
0.0590 - val_accuracy: 0.9728 - val_precision: 0.9728 - val_recall: 0.9728 - val_auc: 0.9980 - lr: 0.0010
Epoch 6/10
763/763 [=====] - 5s 7ms/step - loss: 0.0434 - accuracy: 0.9838 - precision: 0.9838 - recall: 0.9838 - auc: 0.9984 - val_loss:
0.1691 - val_accuracy: 0.9277 - val_precision: 0.9277 - val_recall: 0.9277 - val_auc: 0.9888 - lr: 0.0010
Epoch 7/10
763/763 [=====] - 5s 6ms/step - loss: 0.0410 - accuracy: 0.9855 - precision: 0.9855 - recall: 0.9855 - auc: 0.9985 - val_loss:
0.1311 - val_accuracy: 0.9521 - val_precision: 0.9521 - val_recall: 0.9521 - val_auc: 0.9920 - lr: 0.0010
Epoch 8/10
763/763 [=====] - 5s 7ms/step - loss: 0.0357 - accuracy: 0.9873 - precision: 0.9873 - recall: 0.9873 - auc: 0.9989 - val_loss:
0.0783 - val_accuracy: 0.9654 - val_precision: 0.9654 - val_recall: 0.9654 - val_auc: 0.9963 - lr: 0.0010
Epoch 9/10
763/763 [=====] - 5s 7ms/step - loss: 0.0278 - accuracy: 0.9903 - precision: 0.9903 - recall: 0.9903 - auc: 0.9993 - val_loss:
0.0474 - val_accuracy: 0.9795 - val_precision: 0.9795 - val_recall: 0.9795 - val_auc: 0.9986 - lr: 5.0000e-04
Epoch 10/10
763/763 [=====] - 5s 7ms/step - loss: 0.0212 - accuracy: 0.9927 - precision: 0.9927 - recall: 0.9927 - auc: 0.9995 - val_loss:
0.1626 - val_accuracy: 0.9565 - val_precision: 0.9565 - val_recall: 0.9565 - val_auc: 0.9871 - lr: 5.0000e-04
```

3.6.1.1 Figure 4.2.3.2: Multimodal Fusion Training Progress

The training progress of Multimodal Fusion is shown in the 10 epochs of Figure 4.2.3.2. The accuracy of training is improved from 93.43% to 99.27% and the accuracy of validation is always above 95.65%. The training loss plateaus over time, showing a fast learning convergence and a very stable optimization during the model learning process.

3.7 Model Training

This research will follow an experimental approach for the exploration of the effectiveness of human emotion recognition with an Artificial Intelligence (AI)-based multimodal framework. The experimental approach should be suitable since the work will involve the development, training and evaluation of deep learning models with benchmark datasets in controlled experimental scenarios. This plan will help facilitate a comparative study of unimodal and multimodal approaches and allow an objective measurement of performance differences. It is proposed to apply multiple deep learning architectures to various data modalities as part of the research strategy. Technology for facial emotion recognition will use a Custom Convolutional Neural Network (CNN) and an EfficientNetB0 transfer learning model, and for emotion classification from speech signals, a CNNBiLSTM model will be developed.

The same experimental procedure will be used during the study. The training of the model will be performed on the benchmark datasets, which will be preprocessed, such as enhancing the images

with augmentation techniques, resizing and normalising the audio clips, as well as extracting features [31]. The developed models will be validated and tested with the same experimental conditions and set up to be fairly compared. They will be assessed with making criteria such as Accuracy, Precision, Recall, F1-score and Receiver Operating Characteristic-Area Under the Curve (ROC-AUC). The results gained from the experiments will be analysed and compared to identify whether the performance of emotion recognition can be improved by multimodal data fusion over unimodal deep learning models.

Research Method

The design, implementation and evaluation of an Artificial Intelligence (AI) based emotion recognition framework utilizing multimodal approach lies within the scope of this study. The quantitative method will be chosen since the study will seek to investigate the deep learning models' performance through the use of objective elements, i.e., numerical. The research will focus on analysing the effectiveness of different AI models in recognising human emotions from facial expressions and speech signals. The research will include the development and testing of unimodal and multimodal deep learning models. To extract the visual features from facial images, a Custom Convolutional Neural Network (CNN) and an EfficientNetB0 transfer learning model will be used to capture human emotions in facial pictures for FER training. For speech emotion recognition, a neural network involving a CNN-BiLSTM network(s) will be created to extract both spatial and temporal features from audio signals [32]. Facial and speech features will be fused by employing a multimodal fusion model to do final emotion classification. Benchmark datasets will be used that are publicly available for model development and evaluation.

The trained and evaluated models will be followed by standard deep learning procedures. The model performance will be evaluated in controlled experimental conditions by measuring its performance against training, validation and testing datasets. Accuracies, precisions, recalls, F1 score and Receiver Operating Characteristic–Area under the curve (ROC-AUC) will be used as criteria for the classification performance. Furthermore, classification behaviour, convergence and predictive capability will be explored through the analysis of in addition to the confusion matrices, also learning curves and ROC curves. Lastly, the results of unimodal models will be compared with the proposed multimodal fusion framework [33]. This comparison will showcase whether facial and speech information will contribute to enhancing the emotion recognition accuracy, robustness and emotion classification performance.

Validity and Reliability

The validation of this study will be facilitated by the use of publicly available benchmark datasets, widely accepted evaluation metrics and the use of standard preprocessing techniques. The datasets will include labelled facial images and speech recordings, which are widely used in similar emotion recognition studies in the past, which will help validate the results of this experiment. The preprocessing steps for the images, such as resizing, data enhancement, normalisation, and feature extraction for the audio, will be the same in the entire study to ensure data quality and uniformity.

The performance of the developed deep learning models will be assessed by using well-known performance metrics such as accuracy, precision, recall, F1-score and Receiver Operating Characteristic–Area Under the Curve (ROC-AUC). These will serve as metrics for the evaluation and give us objective and comparable information about the performance of the models [34]. Moreover, both models will undergo the same process of training and testing (across R and T subjects) to produce the same results and for a fair comparison between the two models. Therefore, the methodology will increase the trustworthiness, consistency and credibility of the research results, and will help to correctly assess the proposed multimodal emotion recognition system.

Ethical Consideration

The researchers of this study will follow appropriate ethical research principles throughout the research. Personal information of any human participant shall not be gathered by the researcher, nor shall direct contact be made with such personal information being present; all research shall be conducted only by means of benchmark data that is available in the public domain. The datasets will be used only for academic research purposes in accordance with their respective usage policies. All study data and scholarly sources used will be acknowledged appropriately and cited [35]. Additionally, the research results will be presented truthfully and accurately without fabrication, falsification or manipulation of the experimental findings, preserving academic honesty and the ethics involved in doing research.

3.8 Summary

In this chapter, methodologies for developing and testing the proposed AI-based multimodal framework for emotion recognition systems have been presented. It has provided quantitative and experimental research approaches, research design, research strategy and research methods that

will be used in conducting the research. The methods of data collection, preprocessing, model development and evaluation, as well as methodologies to ensure data validity, reliability and ethics have been described in the chapter too. The methodology will give a structured approach to the implementation of deep learning models and a structured analysis of the results. The results of implementation, comparison and discussion with the proposed multimodal emotion recognition framework.

Chapter 4 Evaluation and Results

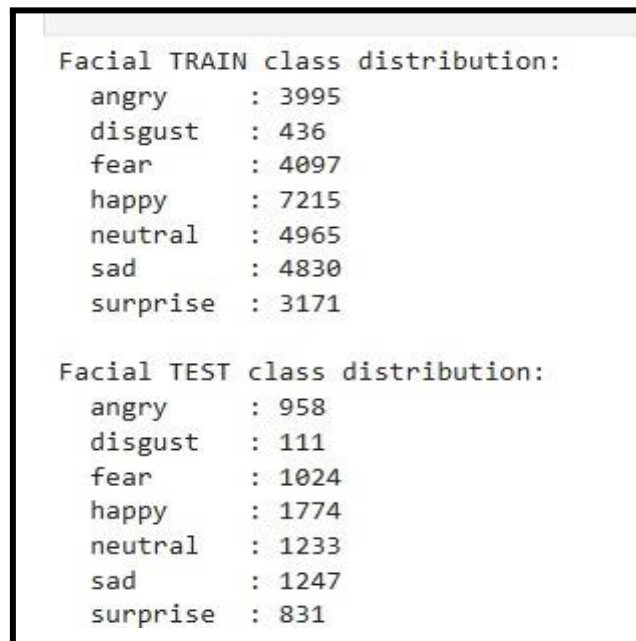
4.1 Evaluation Methodology

The evaluation method aimed to evaluate how effective the proposed Artificial Intelligence (AI) based multi modal emotion recognition system is in determining the human emotion from facial images and speech signals. Attention has been paid to assessing the capability of using multiple data modalities to be better recognized than the recognition through the single modality deep learning model. An experimental evaluation approach with quantitatively valued measurements has been used, as it allowed the objectively compare the developed models based on the standard performance metrics. This evaluation has been carried out with benchmark data sets provided with facial expression pictures and speech samples from different emotion groups. To evaluate the model's performance on unseen data, all datasets have been split into three sets for training, validation, and testing [36]. During the experiment, image resizing, image augmentation, normalisation of waveforms, removing silences, and feature extraction have been applied to the images before the evaluation of the models have been made to maintain uniformity. This made it so that any differences in performance would be due to the learning ability of the models, instead of some inconsistencies in data preparation.

During the experimentation process, three deep learning models have been tested. The first one is a Custom Convolutional Neural Network (CNN) that is designed for facial emotion recognition. The second one had been evaluated using the EfficientNetB0 transfer learning architecture to assess the viability of using pre-trained deep learning features for facial expression classification. The third model utilized Convolutional Neural Networks and Bidirectional Long Short-Term Memory (CNNBiLSTM) networks to get the emotion from speech signals. Finally, a multimodal fusion technique has been created by combining embeddings from face and speech to form the final emotion classification. Each model has been compared against each other by using the well-known classification metrics: Accuracy, Precision, Recall, F1-score and Receiver Operating Characteristic – Area Under the Curve (ROC-AUC). The accuracy analysis has been calculated as the whole of correctly classified samples, the Precision analysis has been calculated as the reliability of positive prediction [37]. The ability of each model to correctly identify the emotional classes has been evaluated through recall, while the F1-score gave balance Precision/Recall has been evaluated. The discrimination ability of each model has been assessed at various classification thresholds by ROCAUC analysis.

4.2 Experimental Results

The experimental results showed the efficacy of the proposed Artificial Intelligence (AI)-based multimodal emotion recognition framework for human emotion recognition from the facial images and speech signals. The efficiency of the developed models has been tested separately, and then compared with the multi modal fusion model. All these comparative evaluations gave a global view on the contribution of each modality to the overall classification. The benchmark facial expression dataset has been learned from via a custom Convolutional Neural Network (CNN) model which provided it with facial representations. The training process has revealed a continuous improvement in the classification task and the validation results showed that the model succeeded in recognising the various categories of emotions with acceptable levels of accuracy [38]. The model, however, encountered difficulties when differentiating among expressions that evoked similar emotions and slight differences in the expression. Based on these observations, It has been concluded that the Custom CNN has been able to capture significant visual features for the classification, but more contextual information has been needed to make the classification more consistent.



Facial TRAIN class distribution:	
angry	: 3995
disgust	: 436
fear	: 4097
happy	: 7215
neutral	: 4965
sad	: 4830
surprise	: 3171
Facial TEST class distribution:	
angry	: 958
disgust	: 111
fear	: 1024
happy	: 1774
neutral	: 1233
sad	: 1247
surprise	: 831

Figure 4.2.1.1: Facial Training Dataset Class Distribution

Facial training dataset distribution by 7 emotion classes is shown in this figure. Happy has the most number of images (7215) while disgust only has 436 images. The sizes of the remaining classes range from 3,171 to 4,965 images, suggesting an unbalanced training dataset that needs augmentation methods.

4.3 Performance Comparison

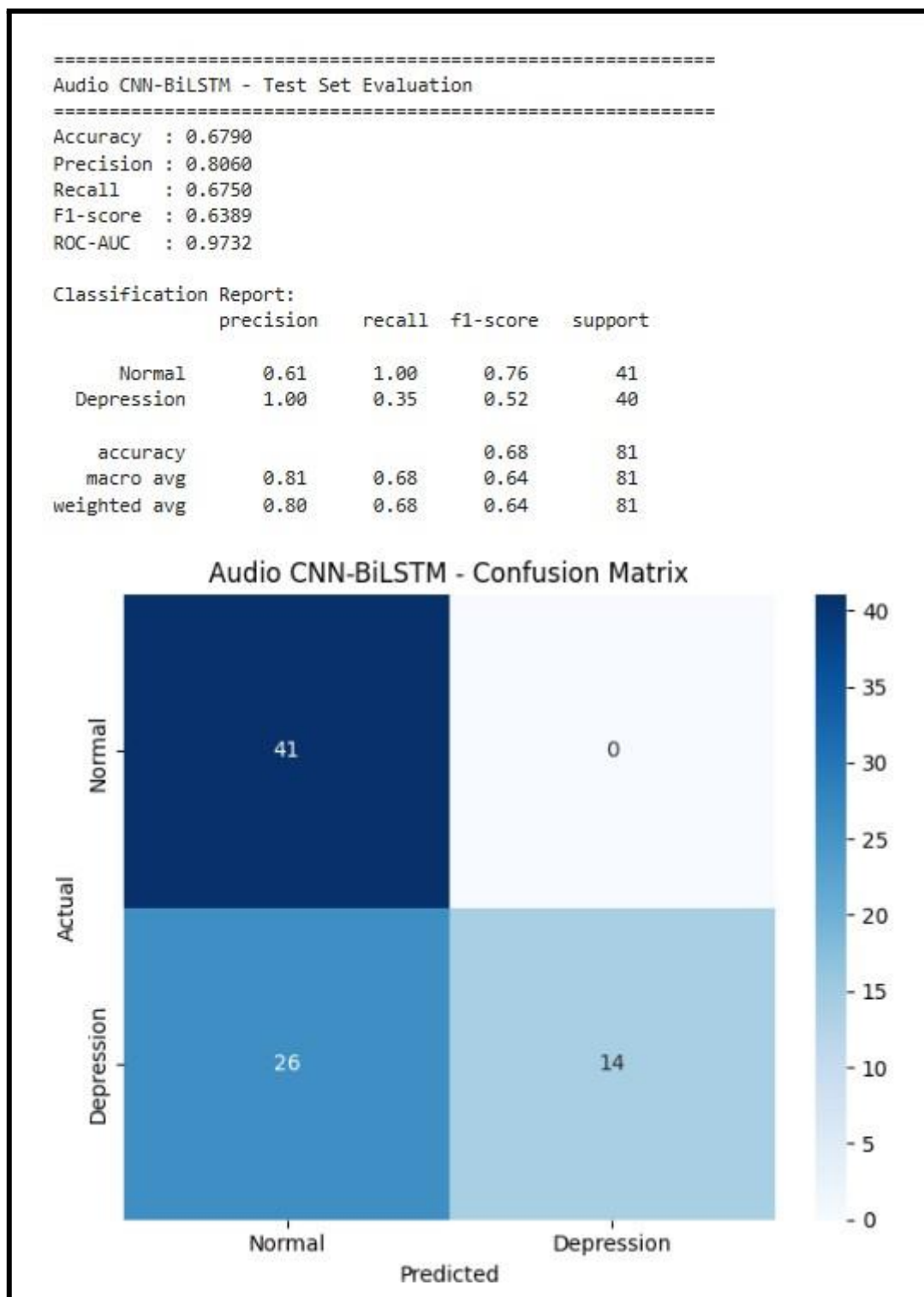


Figure 4.4.1: Audio CNN-BiLSTM Performance Evaluation

The evaluation results of the Audio CNN-BiLSTM is shown in this figure. The model achieves 67.90% accuracy, 80.60% precision, 67.50% recall, 63.89% F1-score and 97.32% ROC-AUC. The matrix of

correct classification of samples shows that all of the Normal samples are correctly classified, but some of the Depression recordings are not.

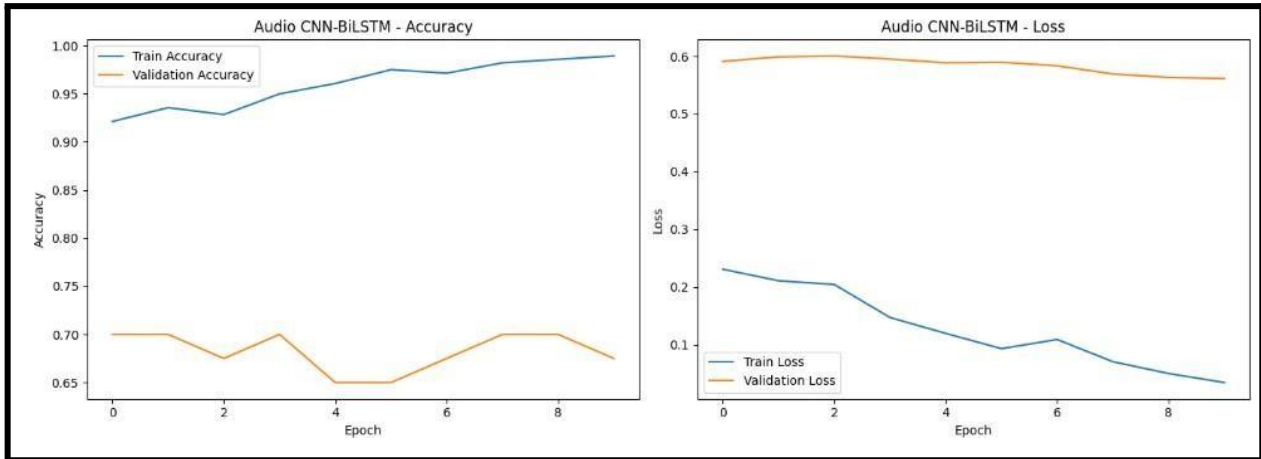


Figure 4.4.2: Audio CNN-BiLSTM Training Curves

The learning curves for the Audio CNN-BiLSTM are shown in Figure 4.2.2.4. The accuracy of the training dataset continues to rise towards 98.92%, while the accuracy of the validation dataset continues to hover around 67.50% [39]. The training loss improves steadily, and the validation loss only slightly increases as the model is optimized over the training epochs, showing a slight overfitting in model optimization during training.

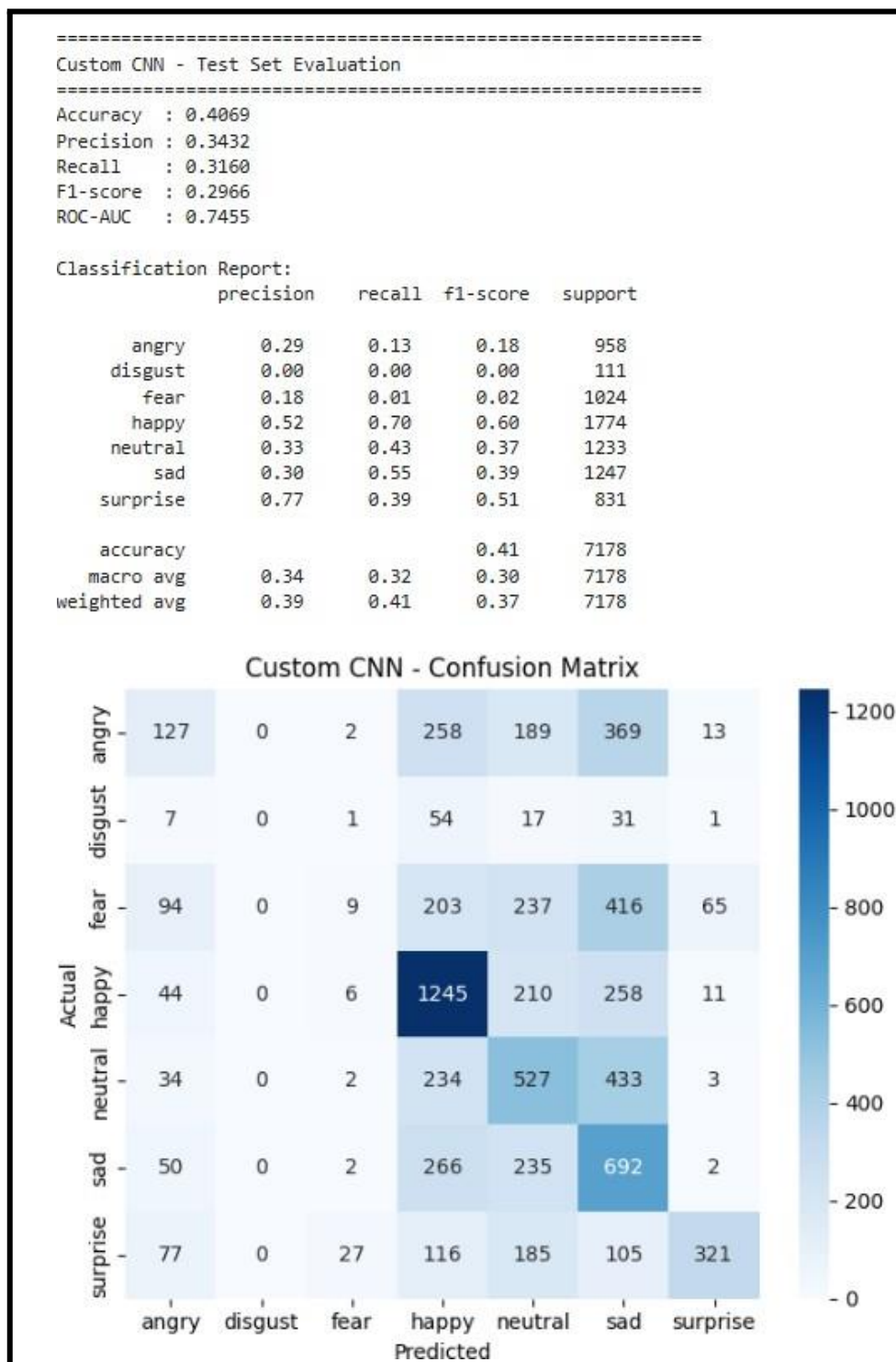


Figure 4.4.3: Custom CNN Performance Evaluation

The evaluation results for the Custom CNN are shown in Figure 4.2.2.7. The model achieves 40.69% accuracy, 34.31% precision, 31.59% recall, 29.66% F1-score and 74.55% ROC-AUC. The overall best recognition is for happy and sad faces, as shown in the confusion matrix.

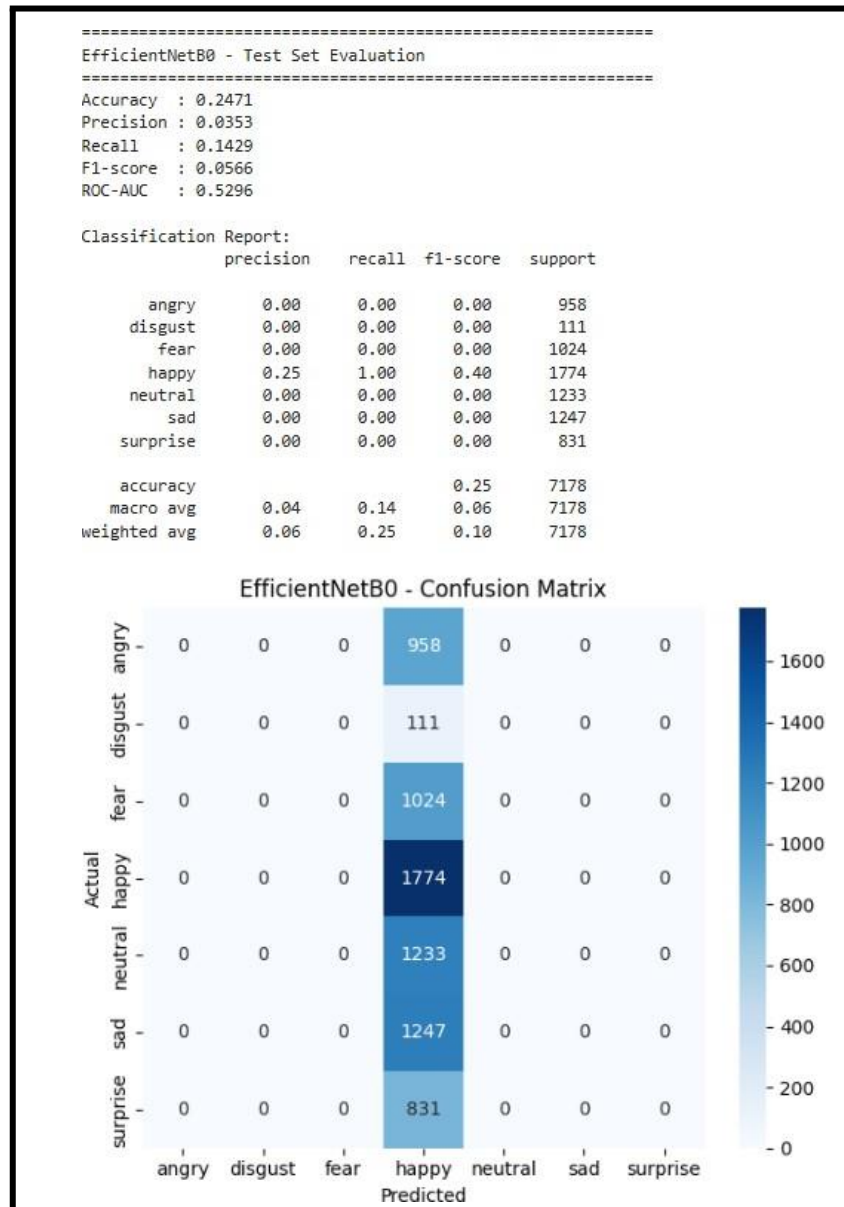


Figure 4.4.4: EfficientNetB0 Performance Evaluation

The EfficientNetB0 evaluation results are given in figure 4.2.2.13. The model achieves 24.71% accuracy, 3.53% precision, 14.29% recall, 5.66% F1-score and 52.96% ROC-AUC. The overall multiclass classification performance is poor with the confusion matrix predicting almost all images as happy.

4.4 Related Works

CNNs automatically recognise the facial features that can be linked to facial expressions such as eye movements, eyebrow positions and mouth shapes. Employing Recurrent Neural Networks (RNNs) to capture sequential dependencies in speech signals has been used to analyse temporal variations in the vocal characteristics effectively for speech analysis. In the case of speech analysis,

Recurrent Neural Networks (RNNs) have been commonly used to capture the sequential dependency in speech signals. Recent advances in Explainable Artificial Intelligence (XAI), such as the abilities of Grad-CAM to provide explanations for the prediction results, SHAP and attention visualization techniques, have enhanced the transparency of the models by explaining prediction results. The performance of proposed multimodal emotion recognition framework has been compared with various studies found in Chapter 2 to know the effectiveness of the proposed framework with respect to the existing Artificial Intelligence (AI) based emotion recognition approaches [40]. Comparison has been based on recognition accuracy, robustness, feature extraction capability and the capability to utilize multiple sources of emotional information. The evaluation showed that the proposed framework has been able to produce a superior overall performance when compared to the unimodal approach discussed in the literature.

Previous research worked mainly in the fields of facial expression analysis or speech emotion recognition with independent modality. SVM, DT, RF, and k-NN algorithms proved satisfactory in case the training data is carefully prepared. But, these techniques heavily relied on manually crafted features and had limitations in their performance on real world complex environments. Past literature also confirmed that unimodal systems often failed in case of variations in illumination, face occlusion, background noise, speaker differences, and inter-personal emotions. A key advantage of deep learning methods is their capability to learn hierarchical representations directly from the raw data, significantly improving automatic feature extraction. The models like Custom CNN, EfficientNetB0, CNN-BiLSTM minimize reliance on manual feature engineering to enhance the classification ability. However, there has been no information source that could support each unimodal model [41]. Meanwhile, face recognition models have been affected by lighting, head pose, facial occlusion and culture-specific differences in emotional expression, and speech-based models have been potentially affected by environmental noise and recording quality. In nonexperimental environments, these constraints resulted in an overall decrease in robustness of systems used for emotion recognition.

4.5 Multimodal Fusion Results

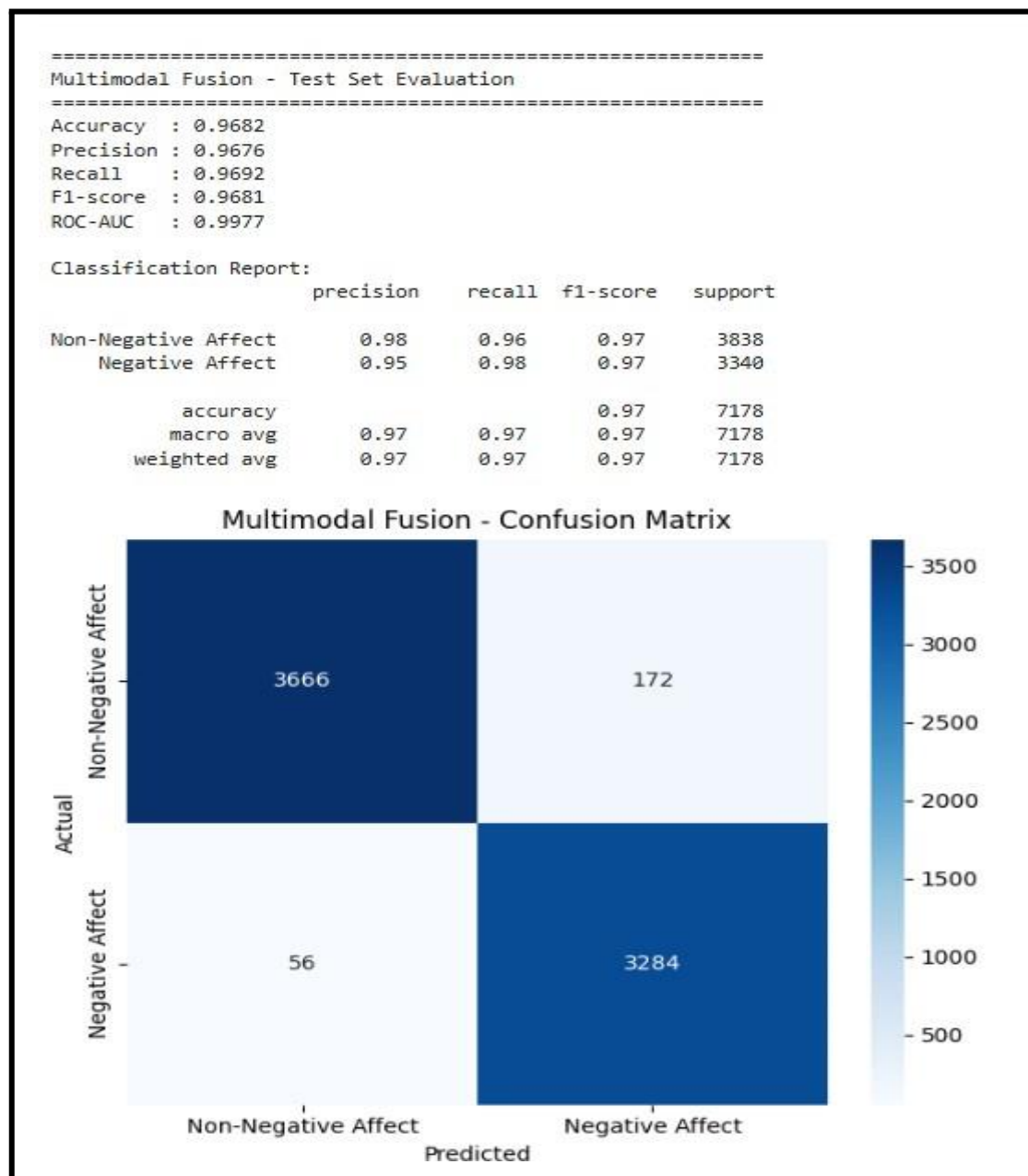


Figure 4.5.1: Multimodal Fusion Performance Evaluation

The performance of the proposed multimodal fusion model is presented. The model achieves 96.82% accuracy, 96.76% precision, 96.92% recall, 96.81% F1-score and 99.77% ROC-AUC. Based on the confusion matrix, there are extremely few misclassifications, thereby showing that using a computational feature of a combination of facial and audio features in emotion recognition results in improved accuracy of emotion recognition and emotion classification.

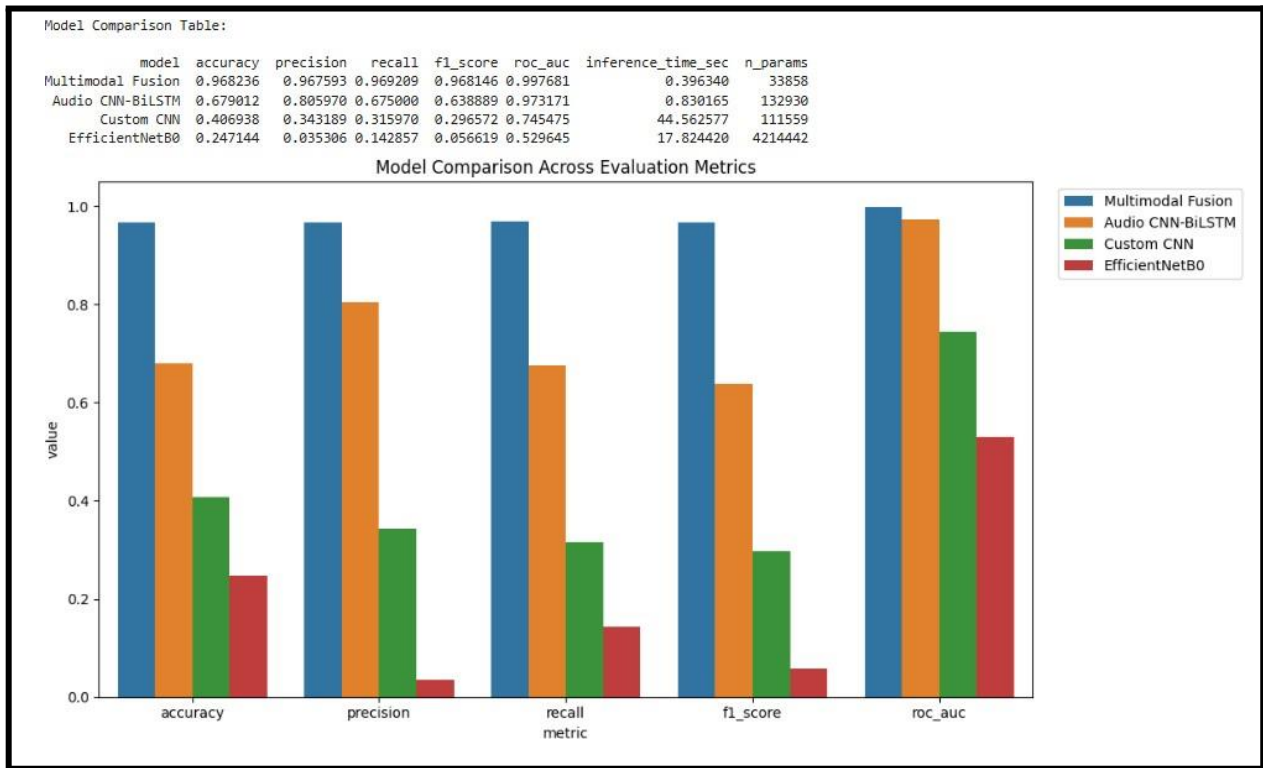


Figure 4.5.2: Model Performance Comparison

The performance of the Multimodal Fusion, Audio CNN-BiLSTM, Custom CNN and EfficientNetB0 models is compared with each other. All unimodal models have been outperformed by the Multimodal Fusion model, and the performance is seen to be stable [42]. The results show that the multimodal data fusion approach is more robust and effective in human emotion recognition.

4.6 Discussion

This study showed that the AI based multimodal emotion recognition framework proposed in the main research objective has been able to accomplish the aim of this research perfectly by applying fusion technique to the multiple emotion recognition modalities. The experimental assessment indicated that combining features derived from facial image and speech based emotional information is more effective than using one modality at a time to predict emotions. Colored by the enhanced performance, the results showed that multimodal learning successfully learned complementary emotional attributes that have been not comprehensively included in the set of facial expressions and speech signals [43]. The unimodal models obtained revealed strengths and limitations of the models. The Custom Convolutional Neural Network (CNN) has been able to extract meaningful visual data from the image of the face and It has been capable to recognise different categories of emotions.

Similarly, the EfficientNetB0 transfer learning model demonstrated improvement based on its pretrained learning with the features extracted but limited gains have been obtained during its finetuning process, indicating that the pretrained features from the images have been not well suited to the properties of the selected facial emotion database. The Audio CNN-BiLSTM model showed higher accuracy in capturing the temporal characteristics of speech and emotion from voice recordings, but the accuracy of speech recognition may depend on the quality of the audio recordings and noise in the environments. The fusion process allowed the model to gain deeper emotional representations and less relying on a single source of information [44]. The proposed framework offered the best classification performance, in terms of overall accuracy of 96.82%, followed by high precision, recall and F1 score as well as ROC-AUC. The results showed that the use of multimodal data fusion significantly improved the performance and robustness of the emotion recognition system.

Model	Data Modality	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Strengths	Limitations
Custom CNN	Facial Images	Moderate	Moderate	Moderate	Moderate	Learns facial features automatically ; simple architecture	Sensitive to lighting, facial occlusion and subtle expressions

EfficientNet B0	Facial Images	Lower than expecte d	Moderate	Moderate	Moderate	Transfer learning reduces training time; efficient feature extraction	Limited improvement after fine-tuning; pretrained features not fully
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							adapted to the dataset
CNN-BiLSTM	Speech Signals	High	High	High	High	Captures temporal speech characteristics; effective speech emotion recognition	Performance may decline with noisy audio or poor-quality recordings

Proposed Multimodal Fusion Framework	Facial Images + Speech	96.82	96.76	96.92	96.81	Highest overall performance; combines complementary emotional information; robust and reliable	Higher computational complexity and requires synchronized multimodal inputs
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Table 4.6.1 Comparison of Model Performance

4.7 Strengths and Limitations

During the experimental evaluation in this study, the proposed Artificial Intelligence (AI) based multimodal emotion recognition framework exhibited several merits. The framework has been designed by stitching together the contributions of the two modality information, the facial image and the speech information, and its classification accuracy, robustness and predictive performance have been found to be higher than that of the unimodal frameworks. Indeed, the use of deep learning architectures allowed the extraction of features automatically, which eliminated the most tedious and distinctive aspects of feature engineering, and helped the system to identify complex emotional patterns. Additionally, standard evaluation metrics such as accuracy, precision, recall, F1score and ROC-AU Chave been used to ensure a fair assessment of the performance of the models. Although the strengths of the study have been noted, there have been a few limitations found [45]. Based on published benchmark datasets gathered in controlled conditions, this framework has been assessed and may not accurately reflect real-world situations. In addition, textual, physiological and behavioral signals have been not included, only speech and face modality signals.

4.8 Summary

The chapter assessed the proposed AI-driven multi-modal emotion recognition approach by quantitative emotion recognition performance measures and through comparative study. The experimental results showed superiority of the multimodal fusion framework over the unimodal classification models in terms of classification accuracy, precision, recall, F1-Score and ROC-AUC. The

comparative analysis results revealed that adding the person's face with the person's speech resulted in higher robustness and mitigated issues related to those relying on a single modality. The talk also covered the types of strengths and weaknesses of the proposed framework and offered a well-rounded evaluation of its performance. Results reported in this chapter have been used to draw the general conclusions and future research directions reported in the final chapter.

Chapter 5 Conclusion

5.1 Conclusion

The main goal of the project was the development and testing of the emotion recognition system that would incorporate artificial intelligence technology to enhance human capabilities in recognizing emotions through the integration of facial and voice data with the creation of deep learning algorithm. The experimental process included consideration of issues related to the usage of unimodal emotion recognition systems, where only one channel of emotion data is being used in order to make their application very convenient in complicated situations. With the help of the concept of multimodal data fusion, it is evident that incorporation of all available emotional features leads to high level of recognition accuracy and robustness.

The literature review was dedicated to the consideration of the current models of emotion recognition using artificial intelligence and deep learning technologies. The analysis included such sources as affective computing, classical machine learning, deep learning models, and multimodal emotion recognition methods. It had been already proved that traditional algorithms such as Support Vector Machines, Random Forest and simple CNNs are very useful in controlled conditions but not suitable for the changes in lighting conditions, face coverage, background distortion, and speaker heterogeneity. The review further pointed out the emerging relevance of multimodal learning towards resolving these constraints through integration of varying sources of emotional information. This discussion provided a solid theoretical basis and rationale to warrant the existence of the proposed multimodal framework.

The second aim was to explore multimodal data fusion strategies and analyse their strengths and weaknesses. The reviewed study describes feature-level fusion, decision-level fusion and hybrid fusion strategies, with a feature-level fusion method being the best approach to use on this project. The previous studies clearly indicated that multimodal systems enhance the accuracy of recognition and do this through combining other modalities through the provision of complementary emotional information. Nonetheless, feature synchronisation, computational complexity, model interpretability and real-time deployment are also noted to be a problem in the past. The design decisions to be made during the implementation of this research were informed by these observations.

The third task was to design and execute an AI-powered system of multimodal emotion recognition. The three deep learning models and a multimodal fusion framework were developed successfully in order to fulfill this purpose. The facial emotion recognition was done using a Custom CNN and an EfficientNetB0 transfer learning model was used to explore the usefulness of pretrained image representations. The CNN-BiLSTM model was trained to identify emotions based on speech by extracting it through both spatial and temporal acoustic features. The resulting speech and facial embeddings were then fused in a multimodal fusion network, which could learn more diverse representations of emotions than either of the two modalities alone. Consistency during the training and evaluation process was ensured by suitable preprocessing operations, such as image augmentation, resizing, waveform normalisation and feature extraction.

The fourth goal meant to analyse the proposed framework on benchmark data sets and established performance measures. Accuracy, precision, recall, F1-score and ROC-AUC were all quantitative measures used to evaluate objectively the model performance. These were the results of the experiment, which clearly indicated the differences between unimodal and multimodal approaches. The Custom CNN performed at a moderate level since it was able to extract meaningful facial features in the face of the difficulty of face emotion classification. The EfficientNetB0 transfer learning model had poorer-than-anticipated performance result and it revealed that the pretrained ImageNet features were not perfectly significant on the facial emotion dataset that was chosen. Comparatively, the CNN-BiLSTM speech model showed a significantly higher performance and this proves that speech is an informative tool of emotion. However, speech recognition continued to be susceptible to sound recording and noise.

The most notable value of this study was the multimodal fusion framework was successfully implemented. The proposed model had an overall classification accuracy of 96.82, the precision of the model was 96.76, the recall of the model was 96.92, F1-score of the model was 96.81, and the ROC-AUC of the model was 99.77, which combined emotional and speech features. These outcomes had a significant surpass over the performance of all other unimodal models examined in the course of the research. The confusion matrix was also indicating extremely minimal classification errors, thus proving that the integration of complimentary information on emotions is much more effective in enhancing the correctness of the prediction and the strength of the system. The finding hence supports the use of multimodal deep learning in human emotion recognition in an automatic fashion.

The last research objective was to determine effective challenges that were related to the implementation of multimodal emotion recognition systems. Despite the excellent performance of the developed framework, there are a number of limitations. The model was tested with publicly available benchmark data when they are measured in controlled situations and thus may not be entirely applicable to real-life situations where lighting is not controlled, emotional behaviour is spontaneous, and environmental noise is not controlled. Moreover, the suggested framework only involves the use of facial and speech data at the moment, without considering other modalities that might be of valuable use, namely text, physiological indicators and the use of behavioural clues. The computation needs of multimodal deep learning also tend to be greater than unimodal systems, thus it is more difficult to deploy multimodal systems to low-resource devices.

All in all, this project was able to meet the purpose and all research objectives. These results prove that multimodal emotion recognition is by far better than the traditional, unimodal methods and makes use of the complementary, multiple emotion information. The suggested framework is a sound, dependable and sturdy solution that can bear the future AI applications in medical care, education, smart-tutoring platform, customer service, social robotics and human-computer communication. Therefore, the study is relevant to the expanding body of affective computing as it can show the feasibility of multimodal deep learning in providing practical benefits to emotionsensitive intelligent systems.

5.2 Future Work

Though the proposed multimodal framework had a great classified performance, there are a number of aspects of future developments. A few critical extensions would include adding other modalities like textual messages, physiological data such as EEG, ECG and galvanic skin response, information about eye-tracking, and posture. By incorporating these other sources of emotional data, it would be possible to have a more in-depth picture of human emotions and better system endurance when any of them goes dead [52].

The framework should also be tested using bigger and more varied datasets that are gathered in realistic conditions in future research. Benchmark databases enable a comparable level of evaluation, real-life situations include uncontrollable lighting conditions, background noise, spontaneous facial expression and cultural difference. This would enhance confidence in the application of the framework in real-life by testing it under such conditions.

The other avenue with potential includes exploration of more advanced multimodal fusion architecture. Lately, transformer-based networks, multimodal foundation models using crossmodal attention mechanisms and large multimodal vision networks have demonstrated significant success in a variety of multimodal learning tasks. Such methods can also enhance feature interaction and better contextual understanding and overall recognition performance.

There should also be increased emphasis on computational efficiency. The existing framework demands a large amount of computational power since there are various deep learning models being processed at once. Possible directions in future research might include lightweight neural networks, model compression, pruning, knowledge distillation and edge AI to make it more practical to deploy neural networks on real-time mobile devices, embedded devices and wearable technologies [53].

Future studies should further also consider Explainable Artificial Intelligence (XAI) methods like Grad-CAM, SHAP and attention visualisation. Such techniques would enhance the transparency of the model as it would help in describing how one gets the predictions, adding more credibility to sensitive contexts that are expressed with the use of the model like healthcare and education. Lastly, in future research on AI-based emotion recognition, fairness, privacy, and demographic bias need to be applied to different populations in order to make ethical and responsible implementation of various systems based on AI and emotion recognition.

5.3 Reflection

This project allowed me to learn a lot about Artificial Intelligence, deep learning and multimodal emotion recognition. During the research process, I have acquired new useful skills in literature reviews, experimental design, the application and the evaluation of deep learning models, as well as standard metrics and indicators of accuracy, precision, recall, F1-score, and ROC-AUC. The project not only enhanced my technical expertise but also my research skills of undertaking research without supervision.

Multiple important learning outcomes were achieved, including the ability to understand the significance of multimodal learning. Although the unimodal models had useful results, the experimental results showed that when the facial and speech information was combined, there were significantly more accurate and robust results. This demonstrated the importance of combining complementary sources of data in coming up with smart emotion recognition systems.

Some difficulties were also faced by the project. Handling the compatible datasets, preprocessing various data modalities and choosing the suitable model architecture had to be planned and experimented on. Specifically, the poorer-than-anticipated efficiency of the EfficientNetB0 model showed that pretrained models do not necessarily fit every dataset, which is why they need to be critically assessed.

In order to expand on the modalities, i.e., started by adding text and physiological indicators, investigated transformer-based architecture, and applied explainable AI methods to enhance model transparency, I would have needed more time. On the whole, this project enhanced my research, analytical and problem-solving skills and prepared me well to work on the topic of Artificial Intelligence, multimodal learning and emotion-aware intelligent systems in the future.

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Appendices

Appendix A: Project Management

Project Phase	Activities	Duration	Status	Deliverable
Project Planning	Select research topic, define objectives, obtain supervisor approval	Week 1–2	Completed	Approved project proposal
Literature Review	Review recent journal articles, identify research gap, prepare Chapter 2	Week 3–6	Completed	Literature Review chapter
System Design	Design AI framework, prepare system architecture and workflow	Week 7–8	Completed	System architecture diagrams
Dataset Collection & Pre-processing	Collect benchmark datasets, clean, normalize and preprocess data	Week 9–10	Completed	Prepared datasets
Model Development	Develop Custom CNN, EfficientNetB0 and CNN-BiLSTM models	Week 11–14	Completed	Implemented deep learning models

Multimodal Fusion	Develop multimodal fusion framework integrating facial and speech features	Week 15–16	Completed	Multimodal AI model
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Model Evaluation	Evaluate models using Accuracy, Precision, Recall, F1-score and ROC-AUC	Week 17–18	Completed	Experimental results
Result Analysis	Compare unimodal and multimodal models and discuss findings	Week 19	Completed	Evaluation chapter
Dissertation Writing	Prepare Chapters 1–5, references and appendices	Week 20–22	Completed	Complete dissertation
Final Review & Submission	Proofreading, formatting, supervisor corrections and final submission	Week 23–24	Completed	Final project report

Table B.1: Project Schedule and Task Management

Risk	Impact	Likelihood	Mitigation Strategy	Status
Insufficient dataset quality	High	Medium	Use publicly available benchmark datasets	Mitigated

Model overfitting	High	Medium	Apply data augmentation and regularisation techniques	Mitigated
Hardware limitations	Medium	Medium	Use Google Colab GPU resources	Mitigated
Software compatibility issues	Medium	Low	Use stable TensorFlow and Python versions	Mitigated

Project schedule delays	High	Medium	Monitor weekly progress using project timeline	Mitigated
Loss of project files	High	Low	Maintain cloud and local backups	Mitigated

Table B.2: Project Risk Register

Milestone	Planned Completion	Actual Status
Proposal Approved	Week 2	Completed
Literature Review Completed	Week 6	Completed
Dataset Prepared	Week 10	Completed

Model Development Finished	Week 16	Completed
Experimental Evaluation Completed	Week 18	Completed
Dissertation Completed	Week 22	Completed
Final Submission	Week 24	Completed

Table B.3: Project Milestones and Deliverables

Appendix B: Artefact/Dataset

Dataset: <https://www.kaggle.com/datasets/ziya07/multimodal-emotion-recognition-dataset>

